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Synaptotagmin 1 and Synaptotagmin 7 promote MR1-mediated presentation of *Mycobacterium tuberculosis* antigens

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eLife Assessment

This **useful** study examines the contribution of synaptotagmin 1 and synaptotagmin 7 to metabolite antigen presentation to mucosal-associated invariant T (MAIT) cells; it begins to address a critical gap in our understanding of the antigen presentation mechanisms of these cells. Strengths of the study include the use of Mtb to study the dynamics of antigen presentation to MAIT cells instead of a synthetic antigen. The strength of the evidence to support the conclusion is **solid**.

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Abstract

Mycobacterium tuberculosis (Mtb) is an intracellular pathogen that can be sensed by T cells, which are essential for the control of infection. In comparison to viral infections, Mtb antigens are relatively limited and hence, challenging to sample. Specialized antigen presentation pathways enable the presentation of such scarce antigens to CD8⁺ T cells, which are, thus, uniquely poised to survey intracellular environments. A subset of CD8⁺ T cells prevalent in the airways, known as mucosal-associated invariant T (MAIT) cells, can be activated through the presentation of Mtb antigens via the MHC class I-related protein 1 (MR1) molecule. Prior work demonstrates that endosomal calcium signaling is critical for MR1-mediated presentation of Mtb-derived antigens. Here, we show that the calcium-sensing trafficking proteins Synaptotagmin (Syt) 1 and Syt7 specifically promote MAIT cell activation in response to Mtb-infected cells. In bronchial epithelial cells, Syt1 and Syt7 localize to late endo-lysosomes and MR1 vesicles. Loss of Syt1 and Syt7 results in enlarged MR1 vesicles and an increased number of MR1 vesicles in close proximity to Mtb-containing vacuoles during infection. This study identifies a specialized pathway in which Syt1 and Syt7 facilitate the translocation of MR1 from Mtb-containing vacuoles, potentially to the cell surface for antigen presentation.

Introduction

Tuberculosis (TB) is the leading cause of infectious disease mortality worldwide with an estimated 10.8 million infections and 1.25 million deaths in 2023¹. TB is caused by the bacillus *Mycobacterium tuberculosis* (Mtb), an intracellular pathogen that invades and replicates within host cells^{2–4}. Unlike viruses, which hijack host machinery to synthesize viral proteins in the cytosol, Mtb antigens are inherently limited. To overcome this limitation, and because Mtb primarily resides within membrane-bound phagosomes, the immune system utilizes multiple pathways to sample distinct subcellular compartments^{5–7}. CD8⁺ T cells, for instance, can recognize Mtb-derived antigens and eliminate infected cells through specialized antigen processing and

presentation mechanisms in myeloid cells as well as the airway epithelium^{2,4,8,9}. Specifically, a subset of CD8⁺ T cells prevalent in the airways, known as mucosal-associated invariant T (MAIT) cells, detects intracellular infections via the major histocompatibility complex (MHC) class I-related protein 1 (MR1) molecule^{10–13}. MAIT cells rapidly respond to infections, produce proinflammatory cytokines upon activation, and promote antimicrobial activity to control infection^{12–14}. Therefore, understanding the specialized processing of Mtb antigens and their presentation by the MR1 molecule could contribute towards the development of effective vaccines and targeted therapies.

CD8⁺ T cells recognize intracellular antigens presented by HLA-Ia and HLA-Ib. While HLA-Ia activates peptide-specific T cells restricted by a specific allele, HLA-Ib presents non-peptide antigens and interacts with donor-unrestricted T cells that recognize a broad spectrum of antigens¹⁵. MR1 is an evolutionarily conserved HLA-Ib molecule widely expressed in nucleated cells^{16,17}. Unlike other HLA-Ib molecules, MR1 presents small molecule metabolites derived from the microbial riboflavin biosynthesis pathway, including those from pathogens such as Mtb, *Salmonella enterica* serovar typhimurium, and *Escherichia coli*^{10,11,18}. Although 5-(2-oxopropylideneamino)-6-d-ribitylaminouracil (5-OP-RU) is the most potent MR1 ligand identified, recent studies show that MR1 presents diverse ligands including those beyond the riboflavin pathway^{19–30}. For example, MAIT cells recognize infection with *Streptococcus pyogenes*, a microbe that cannot produce riboflavin²¹. Additionally, MR1 can present host-derived bile acid metabolites, synthetic ligands, vitamin B6-derived compounds, nucleobase adducts, cigarette smoke components, derivatives of phenylpropanoids, and drug metabolites, leading to MR1-restricted T cell activation^{22,23,25–30}. These findings highlight the diversity and abundance of MR1 ligands beyond what was previously understood.

The processing and presentation of MR1 ligands depend on the intracellular source of antigens. At steady state, MR1 is minimally expressed on the cell surface and predominantly localizes to the endoplasmic reticulum (ER) and late endosomes^{31–33}. Upon binding to small molecule ligands such as 5-OP-RU or acetyl-6-formylpterin (Ac-6-FP), which form a covalent bond with MR1, the ligand-loaded MR1 complex exits the ER and traffics to the cell surface^{32,34}. However, it remains unclear whether this pathway accounts for the processing and presentation of antigens derived from intracellular microbes such as Mtb. Previous work from our group demonstrates the role of endosomal proteins in MR1 trafficking and differences in the presentation of intracellular Mtb antigens compared to exogenous antigens^{31,35,36}. Moreover, recent findings show that inhibition of endosomal calcium release via two-pore channels specifically reduces MR1 presentation of Mtb antigens, but not exogenous antigens³⁷. Thus, endosomal calcium signaling may serve as a key molecular signal in MR1 antigen processing and presentation of intracellular pathogens.

Multiple studies show that local calcium signaling influences vesicular trafficking, including the secretion of lysosome-related organelles and retrograde transport between cellular compartments^{38–40}. As calcium-sensing endosomal trafficking proteins, Synaptotagmins (Syts) are potential downstream effectors of endosomal calcium release that localize to vesicles and mediate membrane fusion events^{41,42}. Syt1, which is highly enriched in neurons, facilitates vesicle exocytosis for neurotransmitter release^{43,44}. In contrast, Syt7 is broadly expressed and localizes to late endosomes and lysosomes in macrophages and dendritic cells, where it mediates the translocation of MHC class II molecules from late endosomes to the plasma membrane^{45–47}. Syts act as part of the soluble N-ethylmaleimide sensitive factor attachment protein receptor (SNARE) complex, which includes vesicle-associated membrane protein (VAMP), Syntaxin, and synaptosome-associated protein. The interaction among these SNARE proteins is essential for membrane fusion and intracellular membrane trafficking^{42,48,49}. These studies suggest that Syts may act as downstream effectors of localized calcium release. The formation of SNARE complexes facilitates vesicle fusion within endosomal compartments or with the plasma membrane, highlighting their potential role in MR1 antigen presentation.

Here, we demonstrate that Syt1 and Syt7 specifically mediate MR1 presentation of Mtb-derived antigens. Loss of Syt1 and Syt7 disrupts cellular distribution and trafficking of MR1 during intracellular Mtb infection, hindering the translocation of the MR1 molecule from Mtb-containing

vacuoles, potentially, to the cell surface for antigen presentation. This study highlights a novel role of Syt1 and Syt7 in MR1 trafficking for the presentation of Mtb antigens and provides critical insights into how the immune system samples and presents intracellular microbes.

Results

Syt1 and Syt7 specifically mediate MR1 presentation of intracellular Mtb in epithelial cells

Based on the known function of Syts in vesicle exocytosis and as calcium sensors^{41,42}, we hypothesized that Syts play a role in MR1-dependent antigen presentation. Among 17 Syt isoforms, Syt1 and Syt7 are the most widely studied Syts⁴⁵. We first defined the expression of Syt1 and Syt7, two calcium-sensing Syts, in human bronchial epithelial BEAS-2B cells as well as differentiated monocytic THP-1 cells. Syt1 and Syt7 transcripts were expressed in both cell types (Fig. 1a⁴⁶). While we previously found that small-interfering RNA (siRNA) knockdown of Syt7 reduced MR1-mediated presentation of Mtb in BEAS-2B cells³⁷, we sought to further define the role of these proteins in the presentation of mycobacterial antigens. As a result, we generated Syt1 and Syt7 knockout (KO) BEAS-2B cells using a lentiviral CRISPR/Cas9 system. We confirmed the gene knockout efficiency in clonal cell lines through Sanger sequencing and used the Inference of CRISPR Editing (ICE) tool to calculate editing efficiency and percent indel distribution⁵⁰. The editing efficiency was 97% for Syt1 with -1 and -21 indel deletions. For Syt7, the editing efficiency was 96% with -14 and -15 indel deletions (Fig. 1b⁴⁷). There was no observed cell toxicity in these clonal cell lines. Then, to determine the functional role of Syt1 and Syt7 in antigen presentation, we measured IFN- γ release from human T cell clones co-cultured with Mtb-infected BEAS-2B cells. We found that both Syt1 and Syt7 KO BEAS-2B cells showed a decrease in MR1 presentation of Mtb (Fig. 1c⁴⁸). To confirm that the degree of intracellular infection was not impacted by genetic deletion of Syt1 and Syt7, we used an HLA-B45-restricted T cell clone specific for the Mtb peptide CFP10₂₋₉. Here, we did not observe any differences in the presentation of Mtb-derived antigen. Also, as controls for exogenously delivered antigens, there were no changes in presentation of *Mycobacterium smegmatis* (Msmeg) supernatant or the CFP10₂₋₉ peptide to the MAIT cell or the HLA-B45-restricted T cell clone, respectively (Fig. 1d⁴⁹). These findings show that genetic deletion of Syt1 and Syt7 specifically inhibited MR1 presentation of intracellular Mtb while leaving HLA-Ia (HLA-B45) presentation and presentation of exogenous antigens intact.

To further investigate if Syt1 and Syt7 contribute to an MR1 endosomal pathway involving recycling endosomes, we used a prodrug of 5-amino-6-D-ribitylaminouracil (5-A-RU) as an antigen⁵¹. 5-A-RU is a precursor of 5-OP-RU, a highly potent MR1 ligand that is unstable without the formation of an MR1-antigen complex. While 5-A-RU is also unstable, a recently developed 5-A-RU prodrug exhibits greater stability. This prodrug requires enzymatic cleavage in acidic compartments for loading to occur in recycling endosomes⁵¹. We found that deletion of Syt1 or Syt7 in BEAS-2B cells did not affect 5-A-RU prodrug presentation (Fig. 1e⁵²). Therefore, Syt1 and Syt7 are selectively involved in MR1 presentation of Mtb-derived antigens discrete from other MR1 endosomal antigen presentation pathways.

Syt1 and Syt7 do not affect Mtb uptake and growth

Prior publications have demonstrated that Syt7 can facilitate phagocytosis in macrophages, but does not affect phagocytic ability in dendritic cells^{46,47}. To examine whether Syt1 and Syt7 play a role in the uptake of Mtb in bronchial epithelial cells, we used an auxotrophic strain of Mtb (AuxMtb). AuxMtb is an attenuated strain of the virulent H37Rv Mtb that retains the region of difference 1 (RD1) locus^{52,53}. RD1 encodes for ESAT-6 and CFP-10, which create pores and lyse phagosomes as in H37Rv Mtb. We further modified AuxMtb into a live/dead reporter strain, mEmeraldRFP-AuxMtb, which constitutively expresses mEmerald and tetracycline-inducible RFP⁵⁴. To assess uptake, we measured the percentage of live GFP⁺ cells after overnight infection (Fig. 2a-b⁵⁵). There were no significant differences in AuxMtb uptake between Syt1 and Syt7 KO and wildtype BEAS-2B cells (Fig. 2c⁵⁶). To determine the viability of intracellular Mtb, we

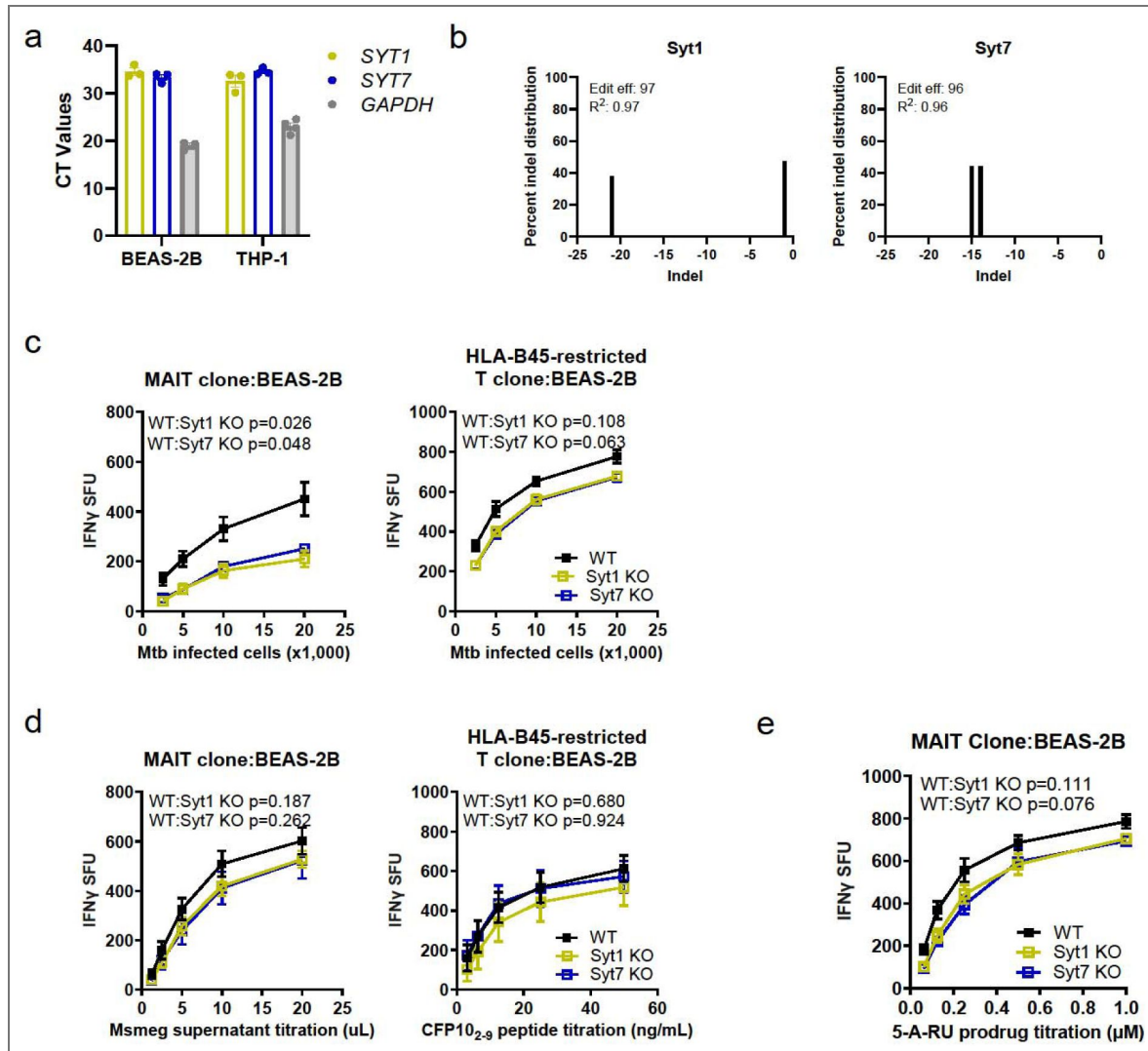


Figure 1. Syt1 and Syt7 specifically mediate MR1 presentation of intracellular Mtb.

(a) Relative gene expression levels of *SYT1*, *SYT7*, and *GAPDH* in BEAS-2B and PMA-differentiated THP-1 cells. (b) Genome editing efficiency and percent distribution of individual indels of Syt1 and Syt7 KO BEAS-2B cells as determined by Sanger sequencing and Interference of CRISPR Edits (ICE)⁵⁰. (c) IFN- γ release by T cell clones (MR1- and HLA-B45-restricted) co-cultured with H37Rv Mtb-infected (MOI=8) WT, Syt1 KO, or Syt7 KO BEAS-2B cells, represented as spot forming units (SFU). (d) IFN- γ release by T cell clones (MR1- and HLA-B45-restricted) co-cultured with WT, Syt1 KO, or Syt7 KO BEAS-2B cells in the presence of Msmeg supernatant, CFP10₂₋₉ peptide, or (e) 5-A-RU prodrug, represented as SFU. All data are plotted as mean $\pm$ SEM and pooled from three independent experiments. For (c-e), the means of technical duplicates were pooled, and non-linear regression analysis of pairwise comparison to WT on best-fit values of top and EC₅₀ were used to calculate p-values by extra sum-of-squares F test.

conducted colony-forming unit (CFU) assays in lysed cells after overnight infection. As shown in Fig.2d [\[link\]](#), we observed no statistical differences in Mtb viability and growth. Lastly, we tested whether there was a change in MR1 expression in Syt1 and Syt7 KO cells and we found no differences in *MR1* transcripts by RT-qPCR compared to wildtype cells (Fig.2e [\[link\]](#)). These findings suggest that Syt1 and Syt7 influence antigen presentation independently of effects on Mtb uptake, Mtb growth, or *MR1* transcript levels.

Syt1 and Syt7 also mediate MR1 presentation of Mtb in THP-1 cells

We next sought to determine whether the roles of Syt1 and Syt7 are generalizable to professional antigen presenting cells. As Syt1 and Syt7 are expressed in differentiated monocytic THP-1 cells (Fig.1a [\[link\]](#)), we generated Syt1 and Syt7 KO THP-1 cells using the CRISPR/Cas9 system. We confirmed knockout efficiency using Sanger sequencing and the ICE analysis tool. The editing efficiency was 95% for Syt1 with -1 and -20 indel deletions and 100% for Syt7 with -16 indel deletion (Fig.3a [\[link\]](#)). To identify the functional role of Syt1 and Syt7 in macrophages for MR1 presentation, we differentiated Syt1 and Syt7 KO THP-1 cells with phorbol 12-myristate 13-acetate (PMA) and measured T cell-dependent IFN- γ release. We found that the genetic deletion of Syt1 and Syt7 in THP-1 cells also resulted in a decrease in MR1 presentation of Mtb, while we observed no changes in the presentation of Msmeg supernatant compared to wildtype cells (Fig.3b-c [\[link\]](#)). Due to HLA mismatch, we were not able to include controls for HLA-Ia and CFP10₂₋₉ peptide. However, these findings indicate that Syt1 and Syt7 also specifically promote MR1 presentation of Mtb and not exogenous antigen in professional antigen presenting cells.

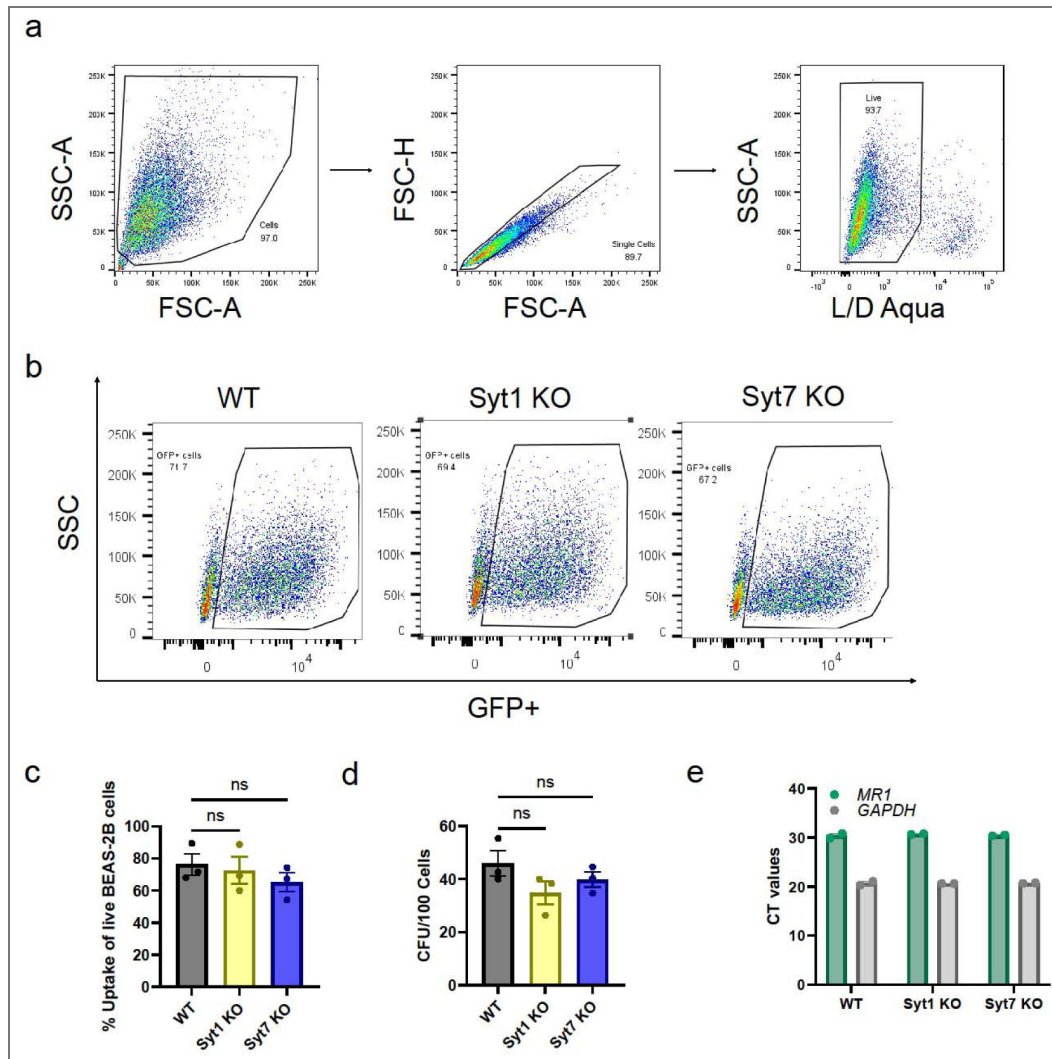


Figure 2. Syt1 and Syt7 do not affect Mtb uptake and growth.

(a) Gating strategy of BEAS-2B cells infected overnight with auxotrophic strain mEmeraldRFP-AuxMtb (MOI=8) by gating on cells, excluding doublets using forward scatter properties, and selecting Live/Dead Near-IR stain negative cells. (b) Representative gate on GFP⁺ population to indicate live BEAS-2B cells infected with mEmeraldRFP-AuxMtb. (c) Percent Mtb uptake measured as proportion of live WT, Syt1 KO, or Syt7 KO BEAS-2B cells that are GFP⁺. (d) Colony forming units (CFU) of H37Rv Mtb in WT, Syt1 KO, or Syt7 KO BEAS-2B cells after overnight infection (MOI=8). (e) Relative gene expression levels of *MR1* and *GAPDH* in WT, Syt1 KO, or Syt7 KO BEAS-2B cells. All data are plotted as mean±SEM. (c-d) are pooled from three independent experiments and (e) is pooled from two independent experiments. For (c-d), ordinary one-way ANOVA with Dunnett's multiple comparisons test were used to analyze significant differences. ns=not significant (p>0.05).

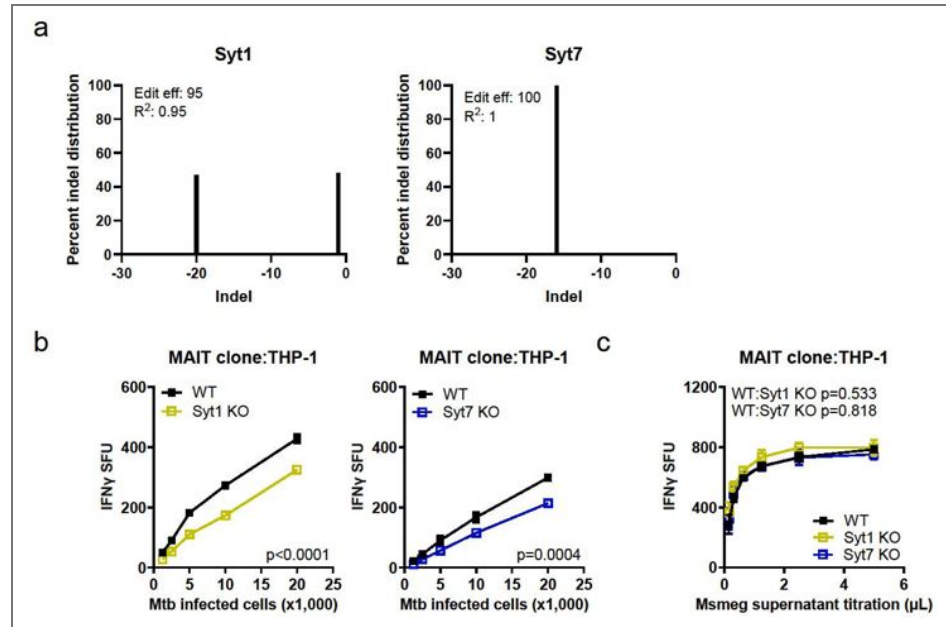


Figure 3. Syt1 and Syt7 also mediate MR1 presentation of Mtb in THP-1 cells.

(a) Genome editing efficiency and percent distribution of individual indels of Syt1 and Syt7 KO THP-1 cells as determined by Sanger sequencing and ICE analysis⁵⁰. (b) IFN- γ release by MAIT cell clones co-cultured with H37Rv Mtb-infected (MOI=1) WT, Syt1 KO, or Syt7 KO THP-1 cells following PMA differentiation, represented as SFU. (c) IFN- γ release by MAIT cell clones co-cultured with WT, Syt1 KO, or Syt7 KO THP-1 cells following PMA differentiation in the presence of Msmeg supernatant, represented as SFU. All data are plotted as mean $\pm$ SEM and pooled from three independent experiments. For (b-c), the means of technical duplicates were pooled, and non-linear regression analysis of pairwise comparison to WT on best-fit values of top and EC₅₀ were used to calculate p-values by extra sum-of-squares F test.

Syt11 and ER-associated Esyt1 and Esyt2 do not selectively affect MR1 presentation of Mtb

To test the hypothesis that Syt1 and Syt7 specifically mediate MR1 presentation of Mtb, we investigated additional calcium-related trafficking proteins^{55,56}. Syt11 is a non-calcium sensing Syt because it does not contain active calcium binding sites⁵⁵. It localizes to recycling endosomes and primarily inhibits endocytosis in neurons⁵⁷. Given the extensive study of the ER pathway in MR1 presentation and the role of the ER as the primary calcium storage site^{32,34,58}, we examined the role of ER-associated calcium-sensing proteins Esyt1 and Esyt2, which facilitate ER-plasma membrane tethering⁵⁹. We found that Syt11, Esyt1, and Esyt2 were highly expressed in BEAS-2B cells (Fig.4a⁶⁰). To determine their roles in MR1 presentation, we transfected BEAS-2B cells with siRNAs targeting *SYT11*, *ESYT1*, or *ESYT2* and confirmed over 80% reduction in mRNA transcripts (Fig.4b⁶⁰). Co-culture with human T cell clones and measurement of T cell-dependent IFN- γ release revealed three distinct phenotypes. Syt11 knock-down increased MR1 presentation of Msmeg supernatant but had no effect on MR1 presentation of Mtb (Fig.4c⁶⁰). In contrast, Esyt2 knockdown reduced MR1 presentation of both Mtb and Msmeg supernatant, while Esyt1 knockdown had no effect (Fig.4d-e⁶⁰). To test whether there is a decrease in MR1 surface translocation in the presence of a ligand in Esyt2 knockdown, we used Ac-6-FP, a potent MR1 ligand that induces MR1 translocation from the ER to the cell surface⁶⁰. Treatment with Ac-6-FP resulted in increased MR1 surface stabilization, but MR1 surface level was significantly lower than those observed in missense control (Fig.S1a-b⁶⁰). Interestingly, the impact of Esyt2 knockdown on MR1 presentation was similar to that of Syntaxin 18 knockdown³¹, both of which localize to the ER

and influence MR1 presentation of exogenous ligands and intracellular Mtb. Moreover, knockdown of these trafficking proteins did not alter HLA-B45 antigen presentation, indicating specific involvement of Syt11 and Esyt2 in MR1 presentation and distinguishing them from traditional HLA-Ia pathways (Fig. 4c-e). These findings demonstrate that among the calcium-related trafficking proteins we investigated, Syt1 and Syt7 uniquely contribute to the MR1 presentation of Mtb without affecting exogenous antigen presentation.

Syt1 and Syt7 localize in late endo-lysosomes and MR1 vesicles

In neurons, Syt1 and Syt7 mainly localize to synaptic vesicles to facilitate membrane fusion^{41,42}. Other studies show that Syt7 is associated with lysosomes in non-neuronal cells^{61,62}. Therefore, we hypothesized that Syt1 and Syt7 will have a similar distribution in lysosomes of bronchial epithelial cells. To define the localization of Syt1 and Syt7, we transfected BEAS-2B cells with RFP-tagged Syt1 and Syt7 and infected them with baculoviruses that deliver constructs to express fluorescent fusion proteins targeted at different subcellular compartments⁶³. Using fluorescence microscopy, we found that both Syt1 and Syt7 highly co-localized with late endosomes (Rab7a; 60.4%±1.7% of vesicles) and lysosomes (LAMP1; 53.0%±2.4% of vesicles) compared to early endosomes (Rab5a; 9.3%±1.3% of vesicles) (Fig. 5a-b). Next, we sought to identify the association of Syt1 and Syt7 with MR1 by live-cell imaging. MR1 localizes in the ER and late endo-lysosomes³¹⁻³³. We used BEAS-2B cells expressing GFP-tagged MR1 to transfect RFP-tagged Syt1 and Syt7. We observed that about 60% of MR1 vesicles co-localized with Syt1 and Syt7 (Fig. 5c-d), which is comparable to the extent of co-localization of MR1 with LAMP1³¹. These findings suggest a potential role of Syt1 and Syt7 in trafficking MR1 vesicles to or from endo-lysosomes.

Absence of Syt1 and Syt7 alters MR1 vesicle size in lysosomal compartments

To investigate the role of Syt1 and Syt7 in MR1 vesicular trafficking and cellular distribution, we generated Syt1 and Syt7 KO in BEAS-2B MR1KO cells stably transduced with a doxycycline-inducible MR1-GFP plasmid (BEAS-2B:TET-MR1GFP)⁶⁴. We validated knockout efficiency using Sanger sequencing and the ICE tool (Fig. 6a). Similar to wildtype BEAS-2B cells, Syt1 and Syt7 KO BEAS-2B:TET-MR1GFP cells exhibited a decrease in MR1 presentation of Mtb (Fig. 6b). Syt7 KO BEAS-2B:TET-MR1GFP cells also demonstrated a small but statistically significant reduction in HLA-B45 presentation of Mtb. However, there was a more pronounced effect in MR1-mediated antigen presentation in Syt7 KO cells compared to WT. We hypothesized that the decreased MR1-mediated presentation of Mtb antigen could be due to decreased MR1 surface translocation in the presence of a ligand. Live-cell imaging showed no differences in MR1 cellular distribution in the presence or absence of Ac-6-FP between WT, Syt1, and Syt7 KO BEAS-2B:TET-MR1GFP cells as MR1 translocated from the ER and vesicles to the cell surface as expected (Fig. 6c). Similarly, Syt1 and Syt7 KO cells did not alter MR1 surface translocation and HLA-Ia expression in the presence or absence of Ac-6-FP when measured by flow cytometry (Fig. 6d). These findings suggest that Syt1 and Syt7 are not involved in the ER pathway of MR1 presentation of exogenous ligands and indicate that the reduction in MR1-mediated presentation of Mtb antigen in Syt1 and Syt7 KO cells is independent of ER-mediated MR1 surface translocation.

We further explored MR1 cellular distribution in Syt1 and Syt7 KO BEAS-2B:TET-MR1GFP cells by live-cell imaging. Interestingly, Syt1 and Syt7 KO cells exhibited enlargement of MR1 vesicles and an accumulation of vesicles in close proximity to one other (Fig. 6e). Quantitative analysis showed a significant increase in the area of larger MR1 vesicles in Syt1 (~60%) and Syt7 (~35%) KO cells compared to WT cells (Fig. 6f). Given that Syt1, Syt7, and MR1 co-localize with markers associated with late endosomes (Rab7a) and lysosomes (LAMP1) at steady state, we hypothesized that these larger and accumulated MR1 collections localized to lysosomal compartments. To test this, we infected Syt1 and Syt7 KO BEAS-2B:TET-MR1GFP cells with baculoviruses expressing RFP in either early endosomes (Rab5a) or lysosomes (LAMP1)⁶³. We found that the larger MR1 vesicles in Syt1 and Syt7 KO cells co-localized more with LAMP1 than Rab5a, similar to WT cells (Fig. 6g-h). These findings suggest that while Syt1 and Syt7 are not involved in the recruitment of MR1

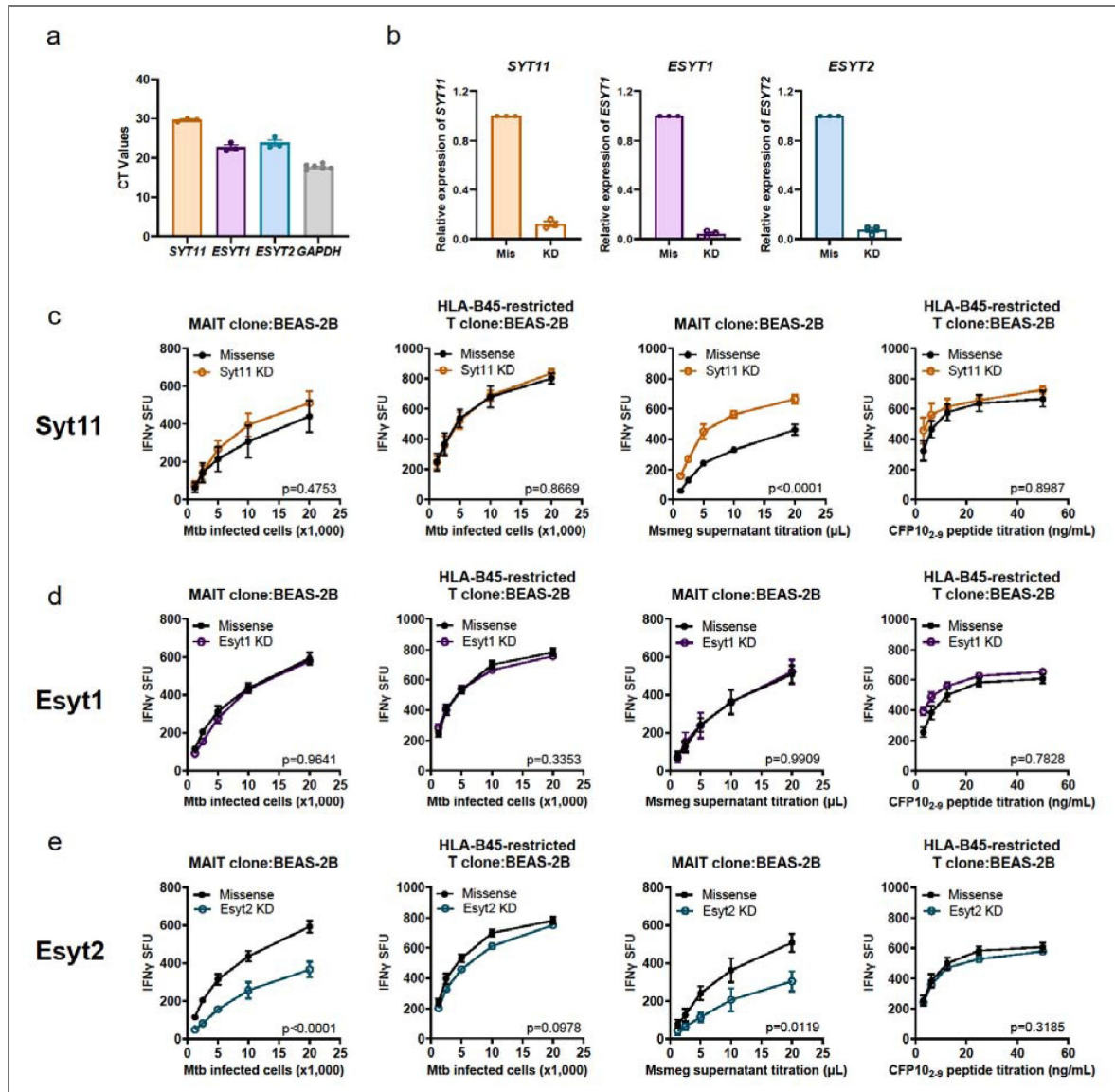


Figure 4. Syt11 and ER-associated Eys1 and Eys2 do not solely affect MR1 presentation of Mtb.

(a) Relative gene expression levels of *SYT11*, *EYS1*, *EYS2*, and *GAPDH* in BEAS-2B cells. (b) Knockdown efficiency of Syt11, Eys1, and Eys2 after 48 hours of knockdown with missense (Mis) or gene-specific (KD) siRNA. (c-e) IFN- γ release by T cell clones (MR1- and HLA-B45-restricted) co-cultured with BEAS-2B cells following siRNA knockdown of Syt11 (c), Eys1 (d), or Eys2 (e). Cells were either infected overnight with H37Rv Mtb (MOI=8) or incubated with exogenously added antigens (Msmeg supernatant and CFP10₂₋₉ peptide). IFN- γ release is represented as SFU. All data are plotted as mean $\pm$ SEM and pooled from three independent experiments. For (c-e), the means of technical duplicates were pooled, and non-linear regression analysis comparing best-fit values of top and EC₅₀ were used to calculate p-values by extra sum-of-squares F test.

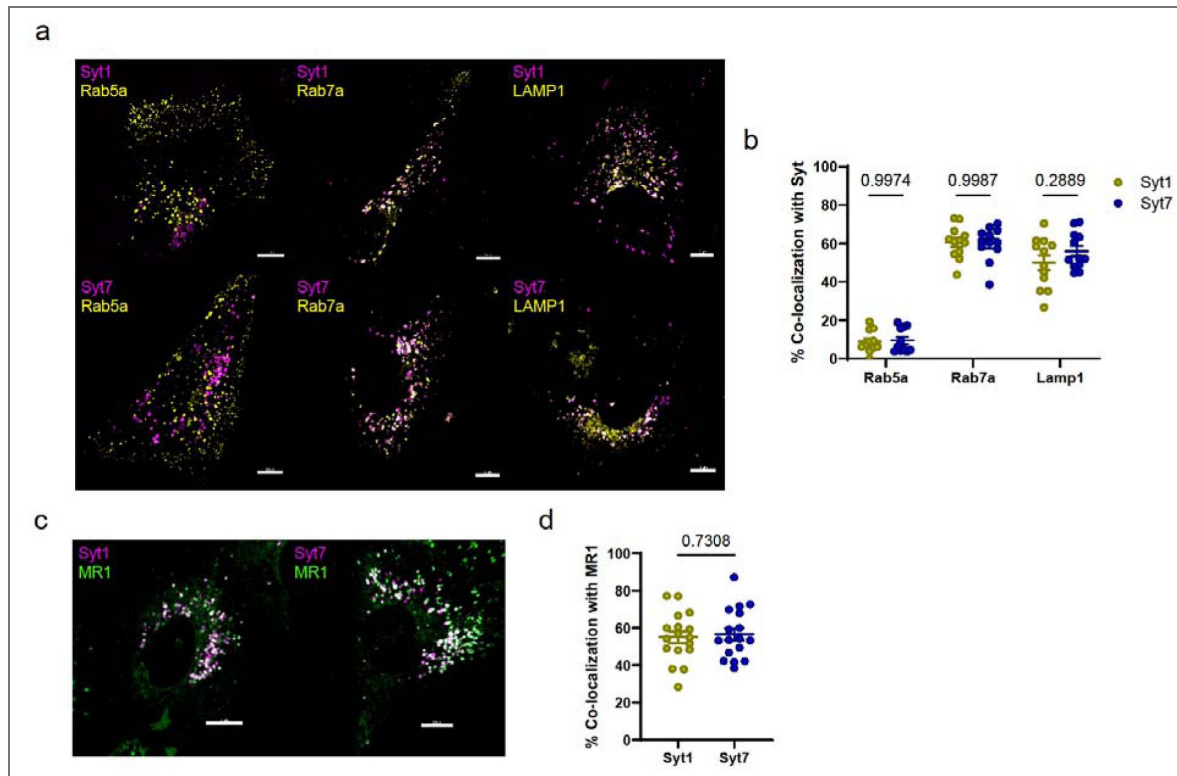


Figure 5. Syt1 and Syt7 localize to late endo-lysosomes and MR1 vesicles.

(a) BEAS-2B cells transfected with Syt1- or Syt7-RFP (magenta) plasmids and incubated with CellLight BacMam 2.0 reagents for Rab5a, Rab7a, and LAMP1 (yellow) overnight. Images are representative of two independent experiments (b) Percent co-localization of Rab5a (n=11), Rab7a (n=12), and LAMP1 (n=12) with Syt1 or Syt7. Data are pooled from two independent experiments and plotted as mean±SEM. Each dot represents one cell. (c) Polyclonal BEAS-2B:TET-MR1GFP (green) cells transfected overnight with Syt1- or Syt7-RFP (magenta) plasmids. Images are representative of three independent experiments. (d) Percent co-localization of Syt1 or Syt7 with MR1 (n=17). Data are pooled from three independent experiments and plotted as mean±SEM. Each dot represents one cell. Two-way ANOVA with Sidak's multiple comparisons test (b) and two-tailed unpaired Student's t-test (d) were used to calculate p-values. All scale bars represent 10 µm.

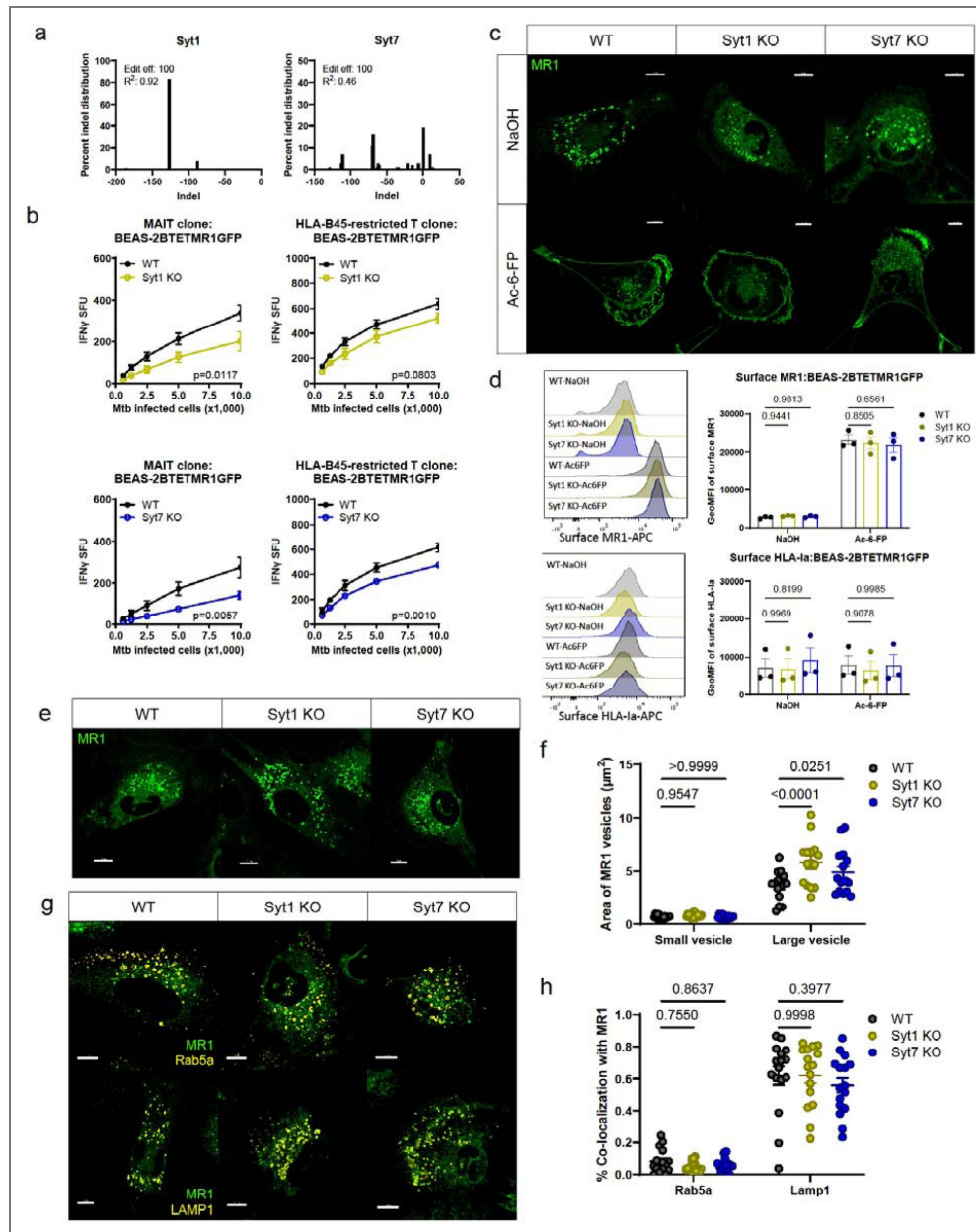


Figure 6. Absence of Syt1 and Syt7 alters MR1 vesicle size in lysosomal compartments.

Syt1 KO and Syt7 KO were generated in the background of BEAS-2B MR1KO:tetMR1-GFP clone D4 cells. (a) Syt1 and Syt7 KO clones were verified by Sanger sequencing and analyzed using the ICE tool⁵⁰. (b) IFN- γ release by T cell clones (MR1- and HLA-B45-restricted) co-cultured with H37Rv Mtb-infected cells (MOI=8) is represented as SFU. The means of technical duplicates were pooled from three independent experiments, and non-linear regression analysis comparing best-fit values of top and EC₅₀ were used to calculate p-values by extra sum-of-squares F test. (c-d) WT, Syt1 KO, and Syt7 KO BEAS-2B MR1KO:tetMR1-GFP cells were incubated overnight with doxycycline and Ac-6-FP or NaOH (solvent control). Images (c) and histograms (d, left) representative of three independent experiments with pooled geometric mean fluorescence intensity (GeoMFI) (d, right) of surface MR1 and HLA-Ia expression. (e) Representative images of WT, Syt1 KO, Syt7 KO BEAS-2B MR1KO:tetMR1-GFP cells incubated overnight with doxycycline and (f) measurement of area of MR1 vesicles, classified into small (1 vesicle) or large (>1 vesicle) (n=15). Each dot represents one cell. Data are pooled from three independent experiments. (g) Representative images of WT, Syt1 KO, Syt7 KO BEAS-2B MR1KO:tetMR1-GFP cells incubated overnight with doxycycline and CellLight BacMam 2.0 reagents for Rab5a and LAMP1 (yellow). (h) Percent co-localization of Rab5a (n=16) and LAMP1 (n=16) with MR1 vesicles. Each dot represents one cell. Data are pooled from four independent experiments. For (d, f, h), p-values were analyzed by two-way ANOVA with Dunnett's multiple comparisons test. All data are plotted as mean $\pm$ SEM. All scale bars represent 10 μ m.

vesicles to LAMP1⁺ compartments, they play a critical role in the trafficking of MR1 vesicles from LAMP1⁺ compartments to other endosomal compartments or to the cell surface for antigen presentation.

Syt1 and Syt7 mediate trafficking of MR1 vesicles from the Mtb-containing vacuole

To understand whether Syt1 and Syt7 are involved in trafficking MR1 vesicles for antigen presentation during intracellular Mtb infection, we conducted live-cell imaging to visualize Mtb-containing vacuoles. Using Syt1 and Syt7 KO BEAS-2B:TET-MR1GFP cells, we infected the cells with wildtype AuxMtb labeled with Alexa Fluor 555 NHS ester dye. Live-cell imaging showed that MR1 vesicles accumulated near Mtb-containing vacuoles in Syt1 and Syt7 KO cells (Fig.7a). Quantitative analysis demonstrated a significant increase in the number of MR1 vesicles within 1 μ m of AuxMtb for Syt1 (1.23 $\pm$ 0.21) and Syt7 KO (1.28 $\pm$ 0.22) cells compared to WT cells (Fig.7b). However, there were no significant differences in the total number or average speed of MR1 vesicles between WT, Syt1, and Syt7 KO cells (Fig.7c). Furthermore, the surface of MR1 vesicles in Syt1 and Syt7 KO cells showed an approximately 4-fold increase in overlap area with Mtb surfaces (Fig.7d). To confirm the increased number and surface overlap of MR1 vesicles with Mtb-containing vacuoles of Syt1 and Syt7 KO cells, we quantified MR1 expression within phagolysosomes using flow organellometry. Our lab previously established this method for separating and phenotyping subcellular compartments^{65,66}. Flow organellometry enables separation of plasma membrane, ER, lysosomes, and phagosomes in a 27% Percoll gradient. Phagosomes are detected in the last few subcellular fractions⁶⁵. We infected WT, Syt1 and Syt7 KO BEAS-2B:TET-MR1GFP cells overnight with wildtype AuxMtb labeled with Alexa Fluor 647 NHS ester dye. We performed flow organellometry to separate phagosomes from other subcellular fractions and identified enrichment of Mtb-containing vacuoles in fractions 42-50 (Fig.7e-f). When comparing MR1 expression within the fractions containing the highest percentage of Mtb⁺LAMP1⁺ vesicles, Syt7 KO cells showed about 25% increase in the geometric mean fluorescence intensity of MR1 compared to WT cells (Fig.7g). Although these findings do not determine the exact location of MR1 loading, they support a model whereby Syt1 and Syt7 transport MR1 vesicles, whether loaded or unloaded, from the Mtb-containing vacuoles and to the cell surface for antigen presentation.

Discussion

CD8⁺ T cells play a unique role in surveilling intracellular environments. Sampling antigens from intracellular microbes is challenging due to their compartmentalization in membrane-bound organelles, low antigen concentrations, and pathogen immune evasion strategies⁵⁻⁸. Thus, the immune system utilizes a specialized mechanism to process and present internalized microbes in the context of HLA-Ia in order to elicit strong cytotoxic T cell responses^{67,68}. Complementing ER-based HLA-Ia-mediated antigen presentation, this highly efficient pathway surveys phagosomes, where foreign antigens are most enriched as they are the first compartment encountered by intracellular microbes. Similarly, HLA-Ib antigen presenting molecules sense and load unique antigens from different subcellular compartments. For example, HLA-E can process and load antigens in the phagosomes, whereas different CD1 isoforms have distinct antigen sources and loading compartments^{65,69}. Interestingly, the known MR1 ligands are small, hydrophilic, and unstable under physiological conditions, which are unlikely to be easily diffused or transported between subcellular compartments^{51,70}. Accordingly, this study highlights a distinct MR1 endosomal antigen presentation pathway mediated by Syt1 and Syt7 during intracellular Mtb infection and suggests MR1 antigen sampling could occur in Mtb-containing vacuoles.

Although the ER pathway of MR1 presentation is well characterized, this pathway does not fully address processing and presentation of intracellular microbes. McWilliam and colleagues show that MR1 mainly resides in the ER where the loading of exogenous ligand occurs. Small ligands such as 5-OP-RU and Ac-6-FP form a Schiff base bond with lysine 43 of MR1, serving as a molecular signal for MR1 to egress the ER and reach the cell surface^{11,32}. Also, treatment with Brefeldin A

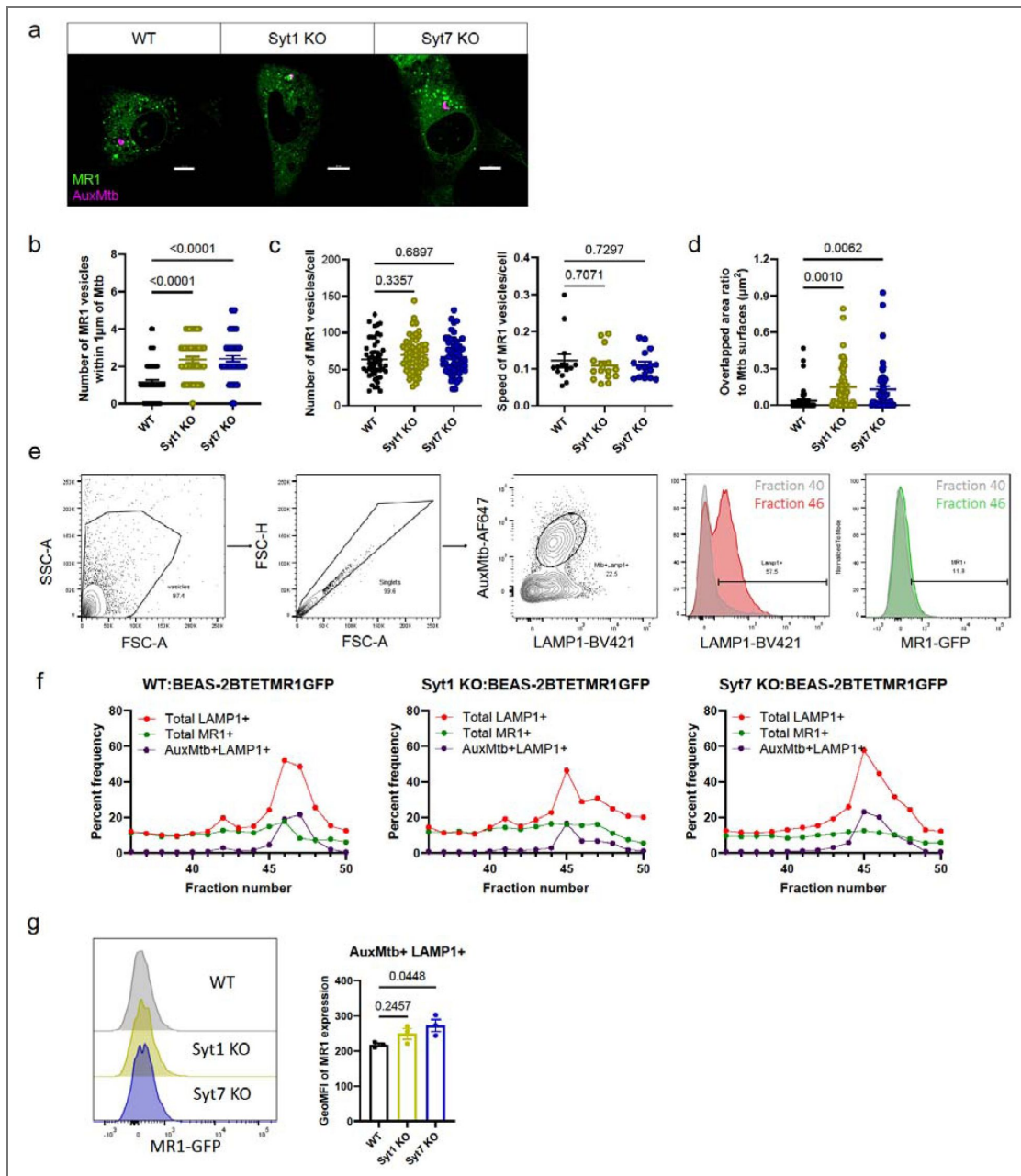


Figure 7. Syt1 and Syt7 mediate trafficking of MR1 vesicles from the Mtb-containing vacuole.

(a) WT, Syt1 KO, Syt7 KO BEAS-2B MR1KO:tetMR1-GFP (green) cells were infected with AuxMtb (MOI=5) labeled with Alexa Fluor 555 Succinimidyl Ester (AuxMtb-Alexa Fluor555; magenta). Images representative of three independent experiments are shown. Scale bars represent 10 μm . (b) Number of MR1 vesicles within 1 μm of the center of the AuxMtb surface ($n=51-53$). (c) Total number (left, $n=51-53$) and average speed ($\mu\text{m/s}$) (right, $n=14-16$) of MR1 vesicles. (d) Overlapped area ratio of MR1 to Mtb surfaces ($n=51-53$). For (b-d), data are plotted as mean $\pm$ SEM and pooled from eight independent experiments. Each dot represents one cell. P-values were analyzed by a one-way ANOVA with Dunnett's multiple comparisons test. (e) WT, Syt1 KO, Syt7 KO BEAS-2B MR1KO:tetMR1-GFP cells were infected overnight with AuxMtb (MOI=10) labeled with Alexa Fluor 647 Succinimidyl Ester. Mtb-containing vacuoles for each fraction from flow organelle assay were identified by gating on vesicles, excluding doublets using forward scatter properties, and selecting the AuxMtb⁺LAMP1⁺ population (left). Total LAMP1⁺ and MR1⁺ populations were gated following the same strategy (right). Representative histograms comparing fractions 40 and 46 are shown. (f) Graphs representative of three independent experiments showing the frequency of total LAMP1⁺, total MR1⁺, and AuxMtb⁺LAMP1⁺ vesicles of all vesicles from fractions 36 to 50. (g) Representative histogram and geometric mean fluorescence intensity (GeoMFI) of MR1 in the subcellular fraction with the highest percentage of AuxMtb⁺LAMP1⁺ vesicles. Data are pooled from three independent experiments and plotted as mean $\pm$ SEM. P-values were analyzed by a one-way ANOVA with Dunnett's multiple comparisons test.

results in diminished recognition of *Salmonella enterica* serovar *typhimurium*, implying transport of loaded MR1 from the ER to the cell surface or MR1 unable to reach the endosomes for loading. Further study has also identified interactions between MR1 and ER-associated chaperone proteins, such as TAPBPR and tapasin³⁴. However, these studies primarily use hematopoietic C1R cells in conjunction with small MR1 ligands added to the extracellular culture medium. Therefore, the mechanisms of MR1 processing and presentation in Mtb-infected antigen presenting cells remain incompletely understood. Further investigation is needed to determine whether this pathway is universal for all intracellular microbes, how it varies across different cell types, and whether specific chaperone proteins assist in antigen loading. These discoveries will extend our understanding of diverse pathways involved in MR1 processing and presentation.

In the endosomal pathway of MR1 presentation, endosomal trafficking proteins and calcium signaling within endosomal compartments facilitate MR1 trafficking and presentation^{31,35–37,71}. Studies investigating the role of endosomal proteins and using exogenously derived antigens and intracellular Mtb infection show multiple MR1 antigen presentation pathways, which are influenced by the type and method of antigen delivery. For instance, VAMP4 knockdown disrupts MR1 presentation of Mtb, while Syntaxin 4 only affects presentation of Msmeg supernatant in bronchial epithelial cells^{31,36}. Additionally, ligand exchange of pre-loaded MR1 occurs in post-ER compartments⁷¹. Moreover, recent evidence shows endosomal calcium signaling as a potential mechanism to detect intracellular infection. Inhibition of endosomal calcium release from two-pore channels specifically decreases MR1 presentation of Mtb in human dendritic cells, setting this pathway apart from other pathways of MR1 antigen presentation³⁷.

As calcium-sensing endosomal trafficking proteins, Syt1 and Syt7 are potential downstream effectors of endosomal calcium release. Syts are membrane-bound proteins in vesicles that mediate docking and fusion to exquisitely regulate exocytosis of synaptic vesicles^{41,42,48}. While Syts are extensively studied in presynaptic neurotransmitter release, recent evidence highlights broader roles for Syt1 and Syt7 in epithelial and antigen presenting cells. Syt1 is expressed in large dense core vesicles of adrenal chromaffin cells, where it regulates hormone release into the bloodstream⁷². Loss of Syt1 in chromaffin cells results in delayed and reduced vesicle exocytosis. Syt7, on the other hand, is broadly expressed across human tissues compared to Syt1⁴⁵. Syt7 localizes to dense lysosomes in rat kidney cells, where its main function is to mediate lysosomal exocytosis⁶¹. Further studies expand the role of Syt7 to immune cells, such as bone marrow-derived macrophages and dendritic cells. In macrophages, Syt7 is essential for efficient phagocytosis and the delivery of lysosomes to nascent phagosomes⁴⁶. In dendritic cells, however, Syt7 deletion does not affect phagocytosis, but significantly delays the translocation of MHC class II to the cell surface⁴⁷. Therefore, we hypothesized that this controlled regulation of exocytosis by Syt1 and Syt7 may play a role in this specialized mechanism of surveilling intracellular microbes by MR1.

The present study supports a unique pathway, in which Syt1 and Syt7 specifically mediate MR1 presentation during intracellular Mtb infection. HLA-Ia presentation of intracellular Mtb as well as presentation of exogenously derived antigens remained intact upon genetic deletion of Syt1 and Syt7. To understand the mechanism, we defined the localization of Syt1 and Syt7 in late endosomes and lysosomes of bronchial epithelial cells. Notably, we did not observe significant differences in Mtb uptake in Syt1 and Syt7 KO cells, similar to previous findings showing that loss of Syt7 does not affect phagocytosis in dendritic cells⁴⁷. However, our Mtb uptake assay involved overnight infections, in which other Syt proteins might compensate for the loss of Syt1 and Syt7 in facilitating phagocytosis. Importantly, the absence of Syt1 and Syt7 in bronchial epithelial cells altered MR1 cellular distribution. Larger MR1 vesicles that accumulated in close proximity to one another were LAMP1⁺, suggesting impaired MR1 vesicle trafficking from lysosomal compartments. This observation aligned with previous studies showing that Syt7 mobilizes lysosomal membrane to other compartments⁴⁶. Although Czibener and colleagues showed that Syt7 is recruited to nascent phagosomes in macrophages during zymosan particle uptake, their follow-up studies demonstrated a different role in dendritic cells. Specifically, in mature dendritic cells, Syt7 translocated to the plasma membrane, and its absence delayed translocation of MHC Class II to the

cell surface⁴⁷. Consistent with these findings, we observed an increase in the number of MR1 vesicles near Mtb-containing vacuoles by fluorescent microscopy and enhanced MR1 expression in Mtb⁺LAMP1⁺ compartments via flow organellometry in Syt1 and Syt7 KO cells. Therefore, we postulate the role of Syt1 and Syt7 in delivering MR1-Mtb antigen complexes from Mtb-containing vacuoles to the cell surface for antigen presentation.

The current work extends our understanding of sampling intracellular microbes through a distinct MR1 endosomal pathway. Inhibition of endosomal calcium release in human dendritic cells and deletion of Syt1 and Syt7 in bronchial epithelial and differentiated monocytic cells all specifically reduce MR1 presentation of Mtb. This suggests that Syt1 and Syt7 act as downstream effectors of endosomal calcium signaling, which facilitate MR1 presentation of Mtb-derived antigens to activate MAIT cells. Moreover, these findings continue to support the existence of distinct MR1 antigen presentation pathways that differ by the type and method of antigen delivery. The endosomal pathway of MR1 presentation involving calcium signaling specifically affects MR1 presentation of Mtb without impacting the ER pathway and other endosomal pathways, reinforcing a specialized mechanism in surveying intracellular microbes. Furthermore, this study expands the role of Syts beyond synaptic vesicle exocytosis, demonstrating their function in antigen presentation during intracellular infection. Our data support a model in which Syt1 and Syt7 mediate the final step in a highly efficient pathway that enables the presentation of MR1-Mtb antigen complexes at the cell surface.

While this study highlights the unique roles of Syt1 and Syt7 in a specialized mechanism of sensing intracellular Mtb, several questions remain. In particular, the potential compensatory roles of other Syt family members and the function of calcium binding domains in antigen presentation warrant further investigation. Follow up studies could identify other SNARE proteins involved in this pathway. Furthermore, our findings suggest that antigen loading may occur in Mtb-containing vacuoles due to their proximity to and the fact that phagosomes are the initial compartments to acquire intracellular microbes. However, additional investigation is needed to track small molecule Mtb-derived antigens in order to determine the exact location of Mtb antigen loading and explore the presence of chaperone proteins in specific subcellular compartments. Overall, this study establishes a foundation for exploring calcium-sensitive endosomal proteins in antigen presentation and its broader implications in sampling intracellular microbes.

Methods

Bacterial strains and cell lines

H37Rv *Mycobacterium tuberculosis* (Mtb) strain, obtained from the American Type Culture Collection (ATCC), was grown in Middlebrook 7H9 Broth supplemented with Middlebrook ADC (BD), 0.05% Tween-80 (OmniPur), and 0.5% glycerol (Thermo Fisher Scientific). Mtb was used from frozen glycerol stocks by passaging 20 times through a tuberculin syringe (BD) with a 27-gauge needle before infection. Multiplicity of Infection (MOI) of 8 was used for IFN- γ ELISpot assays. Δ leuD Δ panCD double auxotroph Mtb (AuxMtb), an attenuated strain of H37Rv Mtb, was gifted by Dr. William Jacobs^{52,53}. A modification of AuxMtb was generated to constitutively express the green fluorescent protein mEmerald and the tetracycline (tet)-inducible red fluorescent protein (mEmeraldRFP-AuxMtb)⁵⁴. MOIs of 5 (for fluorescence microscopy), 8 (for Mtb uptake assay), and 10 (for flow organellometry assay) were used. *Mycobacterium smegmatis* (Msmeg, mc²155 strain) supernatant was obtained by culturing Msmeg for 24 hours in 7H9 broth, passing the supernatant through a 0.22 μ m filter, and concentrating it using a 10 kDa Amicon filter (Millipore Sigma). Msmeg supernatant was used from frozen aliquots stored at -80°C.

BEAS-2B and THP-1 cells were obtained from ATCC. BEAS-2B cells were cultured in DMEM (Gibco) supplemented with 10% heat-inactivated fetal bovine serum (GeminiBio) and L-Glutamine (Gibco). THP-1 cells were cultured in RPMI1640 (Gibco) supplemented with 10% heat-inactivated fetal bovine serum and L-Glutamine. Polyclonal BEAS-2B:tetMR1-GFP cells were developed by transducing BEAS-2B cells with tet-inducible MR1-GFP as described previously³⁵. BEAS-2B MR1KO:tetMR1-GFP clone D4 was generated by reconstituting a clonal BEAS-2B MR1 KO cell line

with a lentiviral vector encoding GFP-tagged MR1A under a tet-inducible promoter⁷³. The polyclonal cell line was sorted based on GFP expression, and individual clonal lines were generated⁶⁴. Due to the constitutive expression of Cas9 and the sgRNA targeting MR1, the MR1 sequence at each insertion of the tetMR1-GFP cassette into the genome was edited. CRISPR editing resulted in a V12I substitution for clone D4, which is phenotypically similar to cell lines over-expressing WT MR1A⁷³. Of note, bovine and porcine MR1 have an I at this position⁷⁴. Clone D4 was then used to generate Syt1 and Syt7 KO BEAS-2B MR1KO:tetMR1-GFP cells.

Lentiviral-mediated gene knockouts

Lentiviral particles for sgRNAs targeting Syt1 (GGTACCATACTCGGAATTGGG) and Syt7 (GGTGTCAGC GCAAACCTGGT) were generated as described previously by co-transfecting low-passage HEK293T cells with lentiviral vector (sgOpti; addgene 85681), packaging vector (psPAX2; addgene 12260), and envelope vector (pVSVg; addgene 138479) using Lipofectamine 2000 (Thermo Fisher)^{75,76}. Generated lentiviral particles were used to transduce low-passage BEAS-2B Cas9 and THP-1 Cas9 cells with 200 µg Polybrene (Sigma). Transduced cells were selected for 6 days with 8 µg/mL Puromycin (Sigma). Monoclonal cell lines were generated by limiting dilution and screened by functional IFN-γ ELISpot assay. Genome editing efficiency and clonality were verified by Sanger sequencing and analyzed with the Inference of CRISPR Edits (ICE, Synthego) tool⁵⁰. Genomic DNA was isolated from KO clones using the QIAmp DNA Mini Kit (Qiagen) per the manufacturer's protocol. DNA surrounding sgRNA target region was amplified with Q5 HF DNA polymerase (NEB) and primers (Syt1 F: CCTCAGTAAGTTACACCCTGAC, Syt1 R: TTGTGTCCAGGGTTGCTGTT, Syt7 F: TGA GGGTCTCAAGTTTGGAGG, Syt7 R: TGTCCCCGAATGTGTCCCTA). Sanger sequencing was performed with an Applied Biosystems 3730xl 96-capillary DNA Analyzer at the OHSU Vollum Institute DNA Sequencing Core.

Ribonucleoprotein-mediated gene knockouts

To generate Syt1 and Syt7 KO in the BEAS-2B MR1KO:tetMR1-GFP clone D4 background, a CRISPR Gene Knockout Kit (Synthego) was used per the manufacturer's protocol on CRISPR editing of immortalized cell lines (Synthego). Ribonucleoprotein (RNP) complexes, consisting of three sgRNAs and SpCas9 nuclease, were assembled at a 9:1 ratio in resuspension buffer R. Electroporation of RNPs was performed with the Neon NxT Electroporation System (Invitrogen) with the following conditions: 1400 V, 20 ms, and 2 pulses. After the electroporated cells rested for 3 days, monoclonal cell lines were generated, and genome editing efficiency was analyzed as described above, with the exception of using the primer sets recommended by Synthego.

Human subjects

All samples were collected with informed consent, and all experiments were conducted according to protocols approved by the Institutional Review Board at Oregon Health & Science University (IRB00000186). Peripheral blood mononuclear cells (PBMCs) were obtained by apheresis from healthy adults and used to expand T cell clones as described below. Human serum was obtained from healthy adults.

T cell clones

Two T cell clones were previously characterized and used throughout the study: a MAIT cell clone (D426-G11)¹⁹ and an HLA-B45-restricted T cell clone (D466-A10)⁷⁷. For rapid expansion, T cell clones were co-cultured with irradiated allogenic PBMCs and allogenic lymphoblastoid cell lines in RPMI1640 medium (Gibco) containing 10% human serum and anti-CD3 (30 ng/mL, clone OKT3, eBioscience)⁷⁸. Recombinant IL-2 (2 ng/mL) was added the following day and every 2-3 days thereafter. T cell clones were washed on Day 5 to remove anti-CD3 and frozen down after at least 11 days. New freeze back stocks were validated before use by comparing IFN-γ response to a previous freeze back.

Reagents and chemicals

5-A-RU prodrug (5-A-RU-PABC-Val-Cit-Fmoc, MedChemExpress) was resuspended in DMSO at 10 mM. Doxycycline (Sigma-Aldrich) was resuspended to 2 mg/mL in H₂O and used at 2 µg/mL unless specified. Ac-6-FP (Schirck's laboratories) was resuspended in 0.01 M NaOH at 5.2 mM. Phytohemagglutinin (PHA, Roche) was resuspended at 10 mg/mL in RPMI1640 medium (Gibco) supplemented with 10% human serum, 2% L-glutamine, and 0.1% gentamicin for ELISpot assays. 16% Paraformaldehyde (PFA, Electron Microscopy Sciences) was diluted to appropriate concentrations. CFP10₂₋₉ peptide was obtained from Genemed Synthesis and resuspended in DMSO at 5 mg/mL.

ELISpot assays

Multiscreen 96-well mixed cellulose esters plates (MSHAS4510, Millipore) were coated overnight at 4°C with 10 µg/mL anti-IFN-γ antibody (Clone 1-D1K, Mabtech) diluted in coating buffer (0.1 M Na₂CO₃, 0.1 M NaHCO₃, pH 9.6). Plates were washed three times with sterile PBS (Corning) and incubated with blocking buffer (RPMI (Gibco) containing 10% human serum, 2% L-glutamine, and 0.1% gentamycin (Gibco)) for 1 hour at room temperature. For exogenous antigens and peptide, 1e4 BEAS-2B cells or 1e4 THP-1 cells differentiated with 50 ng/mL phorbol 12-myristate 13-acetate (PMA) (Sigma-Aldrich) for 48 hours prior were plated in duplicate. Cells were incubated with Msmeg supernatant, CFP10₂₋₉ peptide, or 5-A-RU prodrug in serial dilutions. PHA (Roche) was used as a positive control. After 1 hour of incubation at 37°C and 5% CO₂, 1e4 MAIT cell clone (D426-G11) or 1e4 HLA-B45-restricted T cell clone (D466-A10) were added to the plates based on the antigen. All cells and reagents were resuspended in the blocking buffer described above. Cells were co-cultured for 18 hours at 37°C and 5% CO₂. Plates were washed with PBS (Sigma Aldrich) containing 0.05% Tween-20 (Affymetrix), followed by incubation with ALP-conjugated secondary antibody (Clone 7-B6-1-ALP, Mabtech) diluted in PBS containing 0.5% BSA (Fisher Scientific) and 0.05% Tween-20 for 2 hours at room temperature. Plates were washed again in PBS containing 0.05% Tween-20 and developed using BCIP/NBT-plus developer (Mabtech). IFN-γ spots were enumerated using an AID ELISpot reader and AID EliSpot software (version 7, Autoimmune Diagnostica).

For Mtb infection, 4e5 BEAS-2B cells were infected with H37Rv Mtb (MOI=8) overnight. 4e5 THP-1 cells differentiated with PMA for 48 hours were washed with PBS and infected with H37Rv Mtb (MOI=1) for 3 hours. Subsequently, infected cells were serially diluted and plated in duplicate. 1e4 MAIT cell clone (D426-G11) or 1e4 HLA-B45-restricted T cell clone (D466-A10) were added and co-cultured for 18 hours at 37°C and 5% CO₂. Plates were processed as described above.

CFU assays

Cells infected with H37Rv Mtb (MOI=8) overnight were lysed in ultrapure water the following day. Serial dilutions were plated on 7H10 agar supplemented with glycerol and Middlebrook ADC. Plates were incubated at 37°C and 5% CO₂. Triplicates after serial dilutions with PBS + 0.05% Tween-80 were plated and enumerated after 12 days to determine colony-forming units per 100 cells.

siRNA knockdown

Small-interfering RNAs targeting Syt11 (s23283), Esys1 (s23607), Esys2 (s33136), and a missense control (4390844) were obtained from Thermo Fisher Scientific. 1.5e5 BEAS-2B cells were plated in 6-well plates (Corning) and transfected with 50 nM siRNA using Lipofectamine RNAiMAX (Invitrogen) at 80% confluency. Cells were used for ELISpot assays or RNA extraction at 48 hours post-transfection.

RNA isolation, cDNA synthesis, and qPCR analysis

Total RNA was isolated using the RNeasy Plus Mini Kit (Qiagen) and reverse transcribed into cDNA using the High-Capacity RNA-to-cDNA Kit (Applied Biosystems) according to the manufacturer's instructions. Quantitative RT-qPCR was performed on a Step One Plus Real-Time PCR System (Applied Biosystems) using TaqMan Universal PCR Master Mix (Life Technologies). Taqman FAM-MGB probes for *SYT11* (Hs00383056_m1), *ESYT1* (Hs00248693_m1), *ESYT2* (Hs00294020_m1), *SYT1* (Hs00194572_m1), *SYT7* (Hs01590513_m1), *MR1* (Hs00155420_m1) were obtained from ThermoFisher Scientific. Samples were run in triplicates. Gene expression levels were normalized to *GAPDH* (Hs02758991_g1) of the same sample. Relative expression of the sample was compared to control.

Flow cytometry assays

For Mtb uptake assays, 4e5 BEAS-2B cells were plated in a 6 well plate and incubated for at least 5 hours at 37°C and 5% CO₂. Cells were then infected with mEmeraldRFP-AuxMtb (MOI=8) overnight. Cells were stained with Live/Dead Fixable Dead Cell Stain Kit (Thermo Fisher) for 20 minutes on ice, washed, and fixed in 4% PFA. For MR1 surface expression assays, cells were plated in a 12 well plate and treated with 2 µg/mL doxycycline overnight at 37°C and 5% CO₂. 10 µM Ac-6-FP or NaOH (solvent control) was added the next day for at least 18 hours. Cells were stained with Live/Dead Fixable Dead Cell Stain Kit for 20 minutes on ice. Then, cells were stained with APC-conjugated anti-HLA Ia antibody (clone W6/32; BioLegend #311410), APC-conjugated anti-MR1 antibody (clone 26.5; BioLegend #361108), and APC-conjugated isotype control antibody (clone MOPC-173; BioLegend #400222) for 20 minutes on ice in a FACS buffer (PBS containing 2% human serum, 2% goat serum, and 0.5% FBS). Cells were washed and fixed in 2% PFA. All data was obtained with LSR II (BD) cytometer at the OHSU Flow Cytometry Shared Resource and analyzed with FlowJo software version 10 (TreeStar).

Flow organelle assays

WT, Syt1 KO, and Syt7 KO BEAS-2B MR1KO:tetMR1-GFP cells were plated at 9e6 and incubated with 2 µg/mL doxycycline for 5 hours at 37°C and 5% CO₂. Cells were infected overnight with AuxMtb (MOI=10) labeled with Alexa Fluor 647 Succinimidyl Ester (Invitrogen) in the presence of leucine and pantothenate. Postnuclear supernatant was prepared as described previously⁶⁵. Post-nuclear supernatant was layered onto 27% Percoll and centrifuged for 1 hour at 4°C in 36,000 × g using a 70.1Ti rotor (Beckman). The Percoll gradient was manually fractionated into approximately 50 fractions (200µL each). Fractions 36 to 50, the phagosome-containing fractions, were stained with BV421-conjugated anti-CD107a (LAMP-1) antibody (clone H4A3; BioLegend #328625) in FACS buffer and fixed in 2% PFA. All data was obtained with LSR II (BD) cytometer at the OHSU Flow Cytometry Shared Resource and analyzed with FlowJo software version 10 (TreeStar).

Fluorescence microscopy

All images were acquired on a motorized Nikon TiE stand with a Yokogawa W1 spinning disk unit and a high-powered Agilent laser-emission filter (405nm-445/50nm, 488nm-525/36nm, 561nm-617/73nm). A 100x (NA 1.49) objective was used and images were captured with an Andor Zyla 5.5 sCMOS camera with 2by2 camera binning. For co-localization between Syt1 and Syt7 with other endosomal markers, 1e5 BEAS-2B cells transfected with Syt1-RFP or Syt7-RFP were plated on 8-well #1.5 glass bottom chamber slides (Nunc) and incubated at 37°C and 5% CO₂ overnight. CellLight BacMam 2.0 (Invitrogen) for early endosomes (Rab5a-GFP, C10586), late endosomes (Rab7-GFP, C10588) or lysosomes (LAMP1-GFP, C10596) was added the next day and incubated overnight. For co-localization between Syt1 and Syt7 with MR1, 2-2.5e5 polyclonal BEAS-2B:tetMR1-GFP cells transfected with Syt1-RFP or Syt7-RFP were plated and incubated with 2 µg/mL doxycycline at 37°C and 5% CO₂ overnight. To measure changes in MR1 cellular distribution in Syt1 and Syt7 KO BEAS-2B MR1KO:tetMR1-GFP clone D4 cells, 2e5 cells were plated and incubated with 2 µg/mL doxycycline at 37°C and 5% CO₂. After 4 hours, either CellLight BacMam 2.0 (Invitrogen) for Rab5a-RFP (C10587), Rab7-RFP (C10589), and LAMP1-RFP (C10597) or 10 µM Ac-6-FP and NaOH (solvent

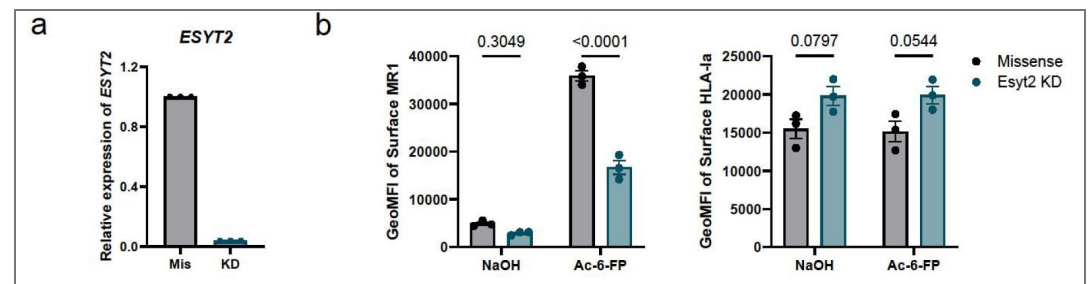
control) were added to quantify co-localization with endosomal compartments or to induce MR1 translocation to the cell surface. For AuxMtb infection, 2-2.5e5 cells were plated on 4-well #1.5 glass bottom chamber slides (Nunc) and incubated with 2 µg/mL doxycycline for 4 hours at 37°C and 5% CO₂. Cells were infected overnight with mEmeraldRFP-AuxMtb or AuxMtb (MOI=5) labeled with Alexa Fluor 555 Succinimidyl Ester (Invitrogen) in the presence of leucine and pantothenate. Before imaging, to ensure randomization cells were stained with or without NucBlue Live ReadyProbes (Invitrogen) and imaged in an unbiased manner based on their NucBlue, RFP, or GFP expression as appropriate.

Image analysis

Co-localizations were analyzed using the “Spots” function and “spots colocalization” MatLab Xtension module on Imaris 7 (Bitplane). Area of MR1 vesicles were classified into small ($\leq 1 \mu\text{m}^2$) and large vesicle ($> 1 \mu\text{m}^2$) using the “Surfaces” function based on MR1-GFP fluorescence, using segmentation setup with Surfaces Detail 0.260 µm, Absolute Intensity Thresholding method, and Classification using surface area. For each cell, areas of small and large MR1 vesicles were averaged in excel. Surface overlaps between MR1 and AuxMtb for each cell were analyzed using the “Surfaces” function similar to above and calculated “overlapped area ratio to surfaces” under detailed specific values using Imaris. Total number and average speed of MR1 vesicles for each cell were analyzed using the “Spots” function. Number of MR1 vesicles within 1 µm of Mtb was enumerated by first defining MR1-GFP vesicles as “Spots” and AuxMtb-RFP as “Surfaces.” Then, MR1 vesicles were classified into two groups ($\leq 1 \mu\text{m}$ and $> 1 \mu\text{m}$) using “shortest distance to surface” with a classification filter. All analysis involving spots, surfaces, and classifications was done using Imaris 7 and 10 (Bitplane).

Statistical analysis

All data were analyzed with Prism 10 (GraphPad). At least three independent experiments were performed and plotted as mean±SEM. Statistical significance for ELISpot assays was determined by non-linear regression using [Agonist] vs. response model with three parameters. Best-fit values of top and EC₅₀ parameters were compared between the curves and p-values were calculated using extra sum-of-squares F test. No constraints were applied to the bottom and top parameters, and the EC₅₀ parameter was constrained to be greater than 0 for p-value calculation. Statistical significances for microscopy and others were determined by two tailed unpaired t-test (for two groups), a one-way ANOVA with Dunnett’s multiple comparisons test (for three groups), or a two-way ANOVA with Sidak’s or Dunnett’s multiple comparisons tests were conducted for statistical analysis. A p value of <0.05 was considered statistically significant.



Supplementary Figure 1. ESYT2 is important for overall MR1 antigen presentation. (a) Knockdown efficiency of ESYT2 after 48 hours of knockdown with missense (Mis) or gene-specific (KD) siRNA. (b) Missense-treated and ESYT2-knockdown cells were incubated overnight with doxycycline and Ac-6-FP or NaOH (solvent control). Histograms representative of three independent experiments with pooled geometric mean fluorescence intensity (GeoMFI) of surface MR1 and HLA-Ia expression. All data are plotted as mean±SEM. For statistics, p-values were analyzed by two-way ANOVA with Sidak’s multiple comparisons test.

Data availability

All relevant data can be found within the article and its supplementary information.

Acknowledgements

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Additional information

Author contributions

The experiments presented were conceptualized by SK, EK, and DML. SK, JCP, AO, and FT performed experiments and analyzed data. EK and DML supervised the work. AO, FT, CK, EK, and DML provided advice and technical expertise. All authors contributed to revising and reviewing the manuscript. All authors approved the final version of the manuscript.

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Additional files

[Supplementary file 1](#) [↗](#)

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Peer reviews

Reviewer #1 (Public review):

Summary:

Synaptotagmin (Syt) 1 and Syt7 specifically promote (are critical for) MAIT cell activation in response to M.tb-infected bronchial epithelial cell line BEAS-2B (Fig. 1) and monocyte-like cell line THP-1 (Fig. 3), but not at the M.smeg-infected conditions. Esyt2 shows a similar effect. This work also displayed co-localization of Syt1 and Syt7 with Rab7a and Lamp1, but not with Rab5a (Fig. 5). Loss of Syt1 and Syt7 resulted in a larger area of MR1 vesicles (Fig. 6f) and an increased number of MR1 vesicles in close proximity to an Auxotrophic Mtb-containing vacuoles during infection (Fig. 7ab). Moreover, flow organellometry to separate phagosomes from other subcellular fractions and identify enrichment of auxotrophic Mtb-containing vacuoles in fractions 42-50, which were enriched with Lamp1+ vacuoles or phagosomes (Fig.7e-f).

Strengths:

This work convincingly associated Syt1 and Syt7 with late endocytic compartments and Mtb+ vacuoles. Gene editing of Syt1 and Syt7 loci of bronchial epithelial and monocyte-like cells supported Syt1 and Syt7 facilitated maintaining a normal level of antigen presentation for MAIT cell activation in Mtb infection. Imaging analyses provided solid evidence to support that Syt1 and Syt7 mutants enhanced the size of MR1-resided vesicles, the overlaps of MR1 with M.tb fluorescent signal, and the MR1 proximity with Mtb-infected vacuoles, suggesting that Syt1 and Syt7 proteins help antigen presentation for MAIT activation in Mtb infection.

Weaknesses:

Current data could be improved to support the conclusion that "This study identifies a pathway in which Syt1 and Syt7 facilitate the translocation of MR1 from Mtb-containing vacuoles, potentially to the cell surface for antigen presentation". Likewise, the current data are more supportive of a different conclusion.

Comments on revisions:

Authors have been very responsive to the review comments, except for keeping a very strong conclusion. Suggest rewriting the conclusions "identifies a specialized pathway", "facilitate the translocation", "from Mtb-containing vacuoles", and "potentially to the cell surface" to be more reflective of the data.

<https://doi.org/10.7554/eLife.108318.2.sa2>

Reviewer #3 (Public review):

Summary:

In the submitted manuscript the authors investigate the role of Synaptotagmins (Syt1) and (Syt7) in MR1 presentation of Mtb antigens. By using Syt1 and Syt7 knock down the authors determine that these molecules are required to effectively control Mtb infection.

Strengths:

In the first series of experiments, the authors determined that knocking down Syt1 and Sy7 in antigen-presenting cells decreases IFN- γ production following cellular infection with Mtb. These experiments are well performed and controlled.

Comments on revisions:

The revised manuscript offers further support to the role of Synaptogamins 1 and 7 in MR1 trafficking during MT infection

<https://doi.org/10.7554/eLife.108318.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Public Reviews:

Reviewer #1 (Public review):

Summary:

The manuscript "Synaptotagmin 1 and Synaptotagmin 7 promote MR1-mediated presentation of Mycobacterium tuberculosis antigens", authored by Kim et al., showed that the calcium-sensing trafficking proteins Synaptotagmin (Syt) 1 and Syt7 specifically promote (are critical for) MAIT cell activation in response to Mtb-infected bronchial epithelial cell line BEAS-2B (Fig. 1) and monocyte-like cell line THP-1 (Figure 3). This work also showed co-localization of Syt1 and Syt7 with Rab7a and Lamp1, but not with Rab5a (Figure 5). Loss of Syt1 and Syt7 resulted in a larger area of MR1 vesicles (Figure 6f) and an increased number of MR1 vesicles in close proximity to an Auxotrophic Mtb-containing vacuoles during infection (Figure 7ab). Moreover, flow organellometry was used to separate phagosomes from other subcellular fractions and identify enrichment of auxotrophic Mtb-containing vacuoles in fractions 42-50, which were enriched with Lamp1+ vacuoles or phagosomes (Figures 7e-f).

Strengths:

This work nicely associated Syt1 and Syt7 with late endocytic compartments and Mtb+ vacuoles. Gene editing of Syt1 and Syt7 loci of bronchial epithelial and monocyte-like cells supported Syt1 and Syt7 facilitated maintaining a normal level of antigen presentation for MAIT cell activation in Mtb infection. Imaging analyses further supported that Syt1 and Syt7 mutants enhanced the overlaps of MR1 with Mtb fluorescence, and the MR1 proximity with Mtb-infected vacuoles, suggesting that Syt1 and Syt7 proteins help antigen presentation in Mtb infection for MAIT activation.

Weaknesses:

Additional data are needed to support the conclusion, "identify a novel pathway in which Syt1 and Syt7 facilitate the translocation of MR1 from Mtb-containing vacuoles" and some pieces of other evidence may be seen by some to contradict this conclusion.

We thank the reviewer for their positive and constructive comments. Because MR1 presents small molecule metabolites, specifically identifying MR1 molecules loaded with antigens derived from intracellular Mtb infection remains a significant technical challenge. Therefore, we agree that some of our approaches measure antigen-loaded MR1 indirectly. For example, IFN- γ release from a MAIT cell clone serves as a sensitive surrogate readout for the presence of antigen-loaded MR1 at the cell surface. This has been demonstrated in previous work showing that IFN- γ release from MAIT cells correlated with loaded MR1 molecules measured using flow cytometry and a TCR based tetramer (Kulicke et al., 2024). In this context, Syt1 and Syt7 represent the first endosomal trafficking proteins we have identified that play a specific role in MR1-mediated presentation of Mtb-derived metabolites. Syt1 and Syt7 do not contribute to the presentation of an exogenously delivered MR1 ligands, such as Ac-6-FP loaded in the ER or *M. smegmatis* supernatant. In Syt1 and Syt7 knockout cells expressing MR1-GFP, larger MR1 vesicles are observed, but MR1 continues to co-localize with LAMP1 similar to wildtype cells. Furthermore, Syt1 and Syt7 knockout cells exhibit an increased number of MR1 vesicles near the Mtb-containing vacuoles compared to wildtype cells. To increase the statistical power of our microscopy analyses, we have analyzed additional cells. Although the absolute magnitude of the observed effects is modest, T cell activation is highly sensitive to the number of loaded antigen presenting molecules at the cell surface. Also, a complementary approach using flow organellometry confirmed increased MR1 expression within Mtb⁺LAMP1⁺ vesicles in Syt7 knockout cells. Thus, these findings suggest a mechanism whereby Syt1 and Syt7 facilitate the trafficking of loaded MR1 molecules from the Mtb-containing vacuoles to the plasma membrane. This specialized mechanism may be analogous to the previously described role of Syt7 in MHC class II trafficking (Becker et al., 2009). In our model, we observed increased accumulation and expression of MR1 within Mtb-containing vacuoles in Syt7 knockout cells.

Reviewer #2 (Public review):

Summary:

The study demonstrates that calcium-sensing trafficking proteins Synaptotagmin (Syt) 1 and Syt7 are involved in the efficient presentation of mycobacterial antigens by MR1 during M. tuberculosis infection. This is achieved by creating antigen-presenting cells in which the Syt1 and Syt7 genes are knocked out. These mutated cell lines show significantly reduced stimulation of MAIT cells, while their stimulation of HLA class I-restricted T cells remains unchanged. Syt1 and Syt7 co-localize in a late endo-lysosomal compartment where MR1 molecules are also located, near M. tuberculosis-containing vacuoles.

Strengths:

This work uncovers a new aspect of how mycobacterial antigens generated during infection are presented. The finding that Syt1 and Syt7 are relevant for final MR1 surface expression and presentation to MR1-restricted T cells is novel and adds valuable information to this process. The experiments include all necessary controls and convincingly validate the role of Syt1 and Syt7. Another key point is that these proteins are essential during infection, but they are not significant when an exogenous synthetic antigen is used in the experiments. This emphasizes the importance of studying infection as a physiological context for antigen presentation to MAIT cells. An additional relevant aspect is that the study reveals the existence of different MR1 antigen presentation pathways, which differ from the endoplasmic reticulum or endosomal pathways that are typical for MHC-presented peptides.

Weaknesses:

The reduced MAIT cell response observed with Syt1 and Syt7-deficient cell lines is statistically significant but not completely abolished. This may suggest that only some MR1-loaded molecules depend on these two Syt proteins. Further research is needed to determine whether, during persistent M. tuberculosis infection, enough MR1-loaded molecules are produced and transported to the plasma membrane to sufficiently stimulate MAIT cells. The study proposes that other Syt proteins might also play a role, as outlined by the authors. However, exploring potential redundant mechanisms that facilitate MR1 loading with antigens remains a challenging task.

We appreciate the reviewer's comments and feedback. Syt1 and Syt7 knockout cells do not completely abolish MR1-mediated presentation of Mtb-derived metabolites. We agree that the likely explanation is that there are redundancies within the antigen presentation pathways. Whether these redundancies are due to other endosomal trafficking proteins or other intracellular compartments where MR1 loading can occur remains unknown. Moreover, Mtb-derived antigens can access the ER, where Syt1 and Syt7 are not involved, thereby enabling an ER-mediated pathway for MR1 antigen presentation. It is also important to note that relatively few (<10) loaded MHC class I molecules are sufficient to trigger T cell activation (Brower et al., 1994; Sykulev et al., 1995; Sykulev et al., 1996). A major challenge in exploring these mechanisms is due to the inability to directly track small molecule Mtb-derived antigens as they are loaded onto MR1 and presented at the cell surface. These hurdles are briefly outlined in the discussion as future directions. Nonetheless, Syt1 and Syt7 are the first endosomal trafficking proteins identified to have a specific effect on MR1-mediated presentation of Mtb-derived antigens.

Reviewer #3 (Public review):

Summary:

In the submitted manuscript, the authors investigate the role of Synaptotagmins (Syt1 and Syt7) in MR1 presentation of MtB.

Strengths:

In the first series of experiments, the authors determined that knocking down Syt1 and Syt7 in antigenpresenting cells decreases IFN- γ production following cellular infection with MtB. These experiments are well performed and controlled.

Weaknesses:

Next, they aim to mechanistically investigate how Syt1 and Syt7 affect MtB presentation. In particular, they focus on MR1, a non-classical MHC-I molecule known to present endogenous and exogenous metabolites, including MtB metabolites. Results from these next series of experiments are less clear. Firstly, they show that knocking down Syt1 and Syt7 does not change MtB phagocytosis as well as MR1 ER-plasma membrane translocation. Based on this, they suggest that Syt1 and Syt7 may affect MR1 trafficking in endosomal compartments. However, neither subcellular compartment analysis nor flow organelleometry clearly establishes the role of Syt1 and Syt7 in MtB trafficking. Altogether, the notion that Synaptotagmins facilitate MR1 interaction with MtB-containing compartments and its vesicular transport was already known. As such, the manuscript should add additional insight on where/how the interaction occurs. The reviewer is left with the notion that Syt1 and Syt7 may affect MR1 presentation, facilitating the trafficking of MR1 vesicles from endosomal compartments to either the cell surface or other endosomal compartments. The analysis is observational and additional data or discussion could address what the insight gained beyond what is already known from the literature.

We thank Reviewer 3 for their comments. Our hypothesis is that Syt1 and Syt7 mediate MR1 trafficking rather than MtB trafficking. While Syt7 has previously been implicated in MHC class II trafficking and vesicular transport, this study is the first to explore in detail the roles of Syt1 and Syt7 in MR1-mediated presentation of MtB-derived metabolites. Since current technologies do not allow direct tracking of MtB-derived antigens loaded onto MR1, we relied on complementary approaches including IFN- γ release from MAIT cells, flow cytometry, fluorescence microscopy, and flow organelleometry. Both flow organelleometry and fluorescence microscopy show increased MR1 expression at MtB-containing vacuoles in Syt7 knockout cells. Since total MR1 expression measured by flow cytometry and the overall number of MR1 vesicles remain unchanged, these data support a mechanism in which Syt7 facilitates the trafficking of antigen-loaded MR1 from MtB-containing vacuoles to the cell surface, consistent with the observed reduction in MAIT cell IFN- γ release.

Recommendations for the authors:

Reviewer #1 (Recommendations for the authors):

Concern 1, the data in the current manuscript have not been sufficient to "identify a novel pathway in which Syt1 and Syt7 facilitate the translocation of MR1 from MtB-containing vacuoles, potentially to the cell surface for antigen presentation" (Last part of Abstract). To conclude this, additional pieces of data are needed: (a) MtB-containing vacuoles associate with MR1 protein expression; (b) MR1+ vesicles traffic from one subcellular location to another; (c) Syt1 or Syt7 KO reduces MR1 vesicles at a downstream subcellular location, e.g., the cell surface. Important evidence supporting the "facilitation

of translocation" is missing on whether Syt1 or Syt7 KO reduces MR1 vesicle traffic from one location to another.

We thank the reviewer for their detailed suggestions to improve our proposed model. We would like to clarify that Figure 7g demonstrates increased MR1 protein expression in Syt7 knockout cells, as assessed by flow organellometry. This approach allowed us to specifically distinguish AuxMtb⁺LAMP1⁺ compartments (Mtb-containing vacuoles) and to quantify MR1 expression using geometric mean fluorescence intensity. Moreover, in both Syt1 and Syt7 knockout cells, MR1⁺ vesicles are retained within lysosomal compartments, characterized by vesicle enlargement and accumulation. Therefore, we did not observe trafficking of MR1⁺ vesicles to other subcellular locations or to the plasma membrane. A key limitation, however, is the lack of current technologies that allow direct measurement of MR1 surface expression specifically during intracellular Mtb infection via flow cytometry. Given this limitation, IFN- γ ELISpot is a sensitive surrogate and supports the conclusion that loss of Syt1 and Syt7 results in decreased MR1 presentation of Mtb-derived antigens at the plasma membrane.

The results "a significant increase in the number of MR1 vesicles within 1 μ m of AuxMtb for Syt1 (1.13 {plus minus} 0.46) and Syt7 KO (1.31 {plus minus} 0.46) cells compared to WT cells (Fig.7b)." and "the surface of MR1 vesicles in Syt1 and Syt7 KO cells showed a 3-fold increase in overlap area with Mtb surfaces (Fig.7d)." may need to be further elaborated on whether MR1+vacuoles and Mtb+ vacuoles are overlapped or are adjacent. Figure 7b shows several groups of vacuoles with the same distance. This needs a larger sample size to randomize this distance measurement, for example, calculating 50~100 Mtb+ vacuoles.

We appreciate the reviewer's critical comments and suggestions. To quantify distance and surface overlap, the microscopy images were acquired from a single optical plane rather than full z-stacks. As a result, it is not possible to definitively determine whether MR1⁺ vesicles and Mtb-containing vacuoles are directly overlapping or adjacent. In response to the reviewer's suggestion, we increased the sample size for both distance (n=51-53) and surface overlap analyses (n=51-53). Using the larger sample size, we observed a significant increase in the number of MR1 vesicles located within 1 μ m of AuxMtb in both Syt1 (1.23 $\pm$ 0.21) and Syt7 knockout (1.28 $\pm$ 0.22) cells. Also, there was an approximately 4-fold increase in MR1-Mtb surface overlap area compared to wildtype cells.

Results from "performed flow organellometry to separate phagosomes from other subcellular fractions and identified enrichment of Mtb-containing vacuoles in fractions 42-50 (Fig.7e-f)" could not distinguish the difference between WT and Syt1/Syt7 KO, or further support the role of Syt1/Syt7 in endocytic trafficking. More specifically, authors claimed that "enhanced MR1 expression in Mtb+LAMP1+ compartments via flow organellometry in Syt1 and Syt7 KO cells.", may not be supported by Figure 7f, which does not show a difference in MR1 expression between Syt1 KO or Syt7 KO and WT.

We appreciate the reviewer's concerns and would like to clarify the interpretation of Figures 7f and 7g. Figure 7f demonstrates: (a) enrichment Mtb-containing vacuoles within fractions 42-50, (b) coenrichment of LAMP1⁺ vesicles within these Mtb-containing fractions, and (c) comparable subcellular fractionation profiles across wildtype, Syt1 knockout, and Syt7 knockout cells, indicating no major differences in fraction distribution. Differences in MR1 expression are shown in Figure 7g, which compares MR1 expression as the geometric mean fluorescence intensity within the fraction exhibiting the highest percentage of AuxMtb⁺LAMP1⁺ across all fractions. We observed significant increase in MR1 expression in Syt7 knockout cells compared to wildtype cells.

Concern 2, in abstract, "Loss of Syt1 and Syt7 results in enlarged MR1 vesicles and an increased number of MR1 vesicles in close proximity to Mtb-containing vacuoles during infection.". Although numbers of MR1 vesicles within 1 μ m of Mtb increase (Figure 7b) and

areas of MR1+ vacuoles for WT and KO cells enhance (Figure 6f), but numbers of MR1 vesicles/cells are not different between WT and Syt1 and Syt7 KO (Fig. 7c). These imaging analyses, including other figure panels, need more explicit presentation of (most if not all) random images for calculation, annotation of MR1-vacuoles for calculation, and raw statistical data for mean and p value calculation. These raw data can be presented in supplemental figure panels.

We thank the reviewer for these suggestions. We have included more details on randomization, technical procedures, and statistical analyses in methods section for “Fluorescence Microscopy,” “Image Analysis,” and “Statistical Analysis.” Raw data collection and statistical data are presented in the supplemental data.

Concern 3, additional evidence that does not support the conclusion "This study identifies a novel pathway in which Syt1 and Syt7 facilitate the translocation of MR1 from Mtb-containing vacuoles" (the last part of Abstract). This additional unsupportive evidence includes: (a) MR1 expression on the cell surface is not impacted or not different among WT, Syt1 KO, and Syt7 KO of BEAS-2B cells (Fig. 6d). (b) "Live-cell imaging showed no differences in MR1 cellular distribution in the presence or absence of Ac-6-FP between WT, Syt1, and Syt7 KO BEAS-2B:TET-MR1GFP cells as MR1 translocated from the ER and vesicles to the cell surface as expected (Figure 6c).

We thank the reviewer for this comment and would like to clarify our use of Ac-6-FP. Figures 6c and 6d examine MR1 cellular distribution and surface expression in the presence or absence of Ac-6-FP. Ac-6-FP is a small MR1 ligand that is loaded in the ER and promotes MR1 surface stabilization and trafficking to the cell membrane. In contrast, Mtb primarily resides within membrane-bound phagosomes. MR1 presentations of soluble/exogenously delivered ligands versus intracellular Mtb-derived antigens have shown to involve distinct pathways and endosomal trafficking proteins (Harrieff et al., 2016; Karamooz et al., 2019; Karamooz et al., 2025). Findings from Figures 6c and 6d show that Syt1 and Syt7 do not contribute to the presentation of small soluble and ER-loaded ligands such as Ac-6-FP. Instead, they specifically contribute in MR1 presentation of Mtb-derived metabolites by translocating MR1 from Mtb-containing vacuoles in the context of intracellular Mtb infection

Other concerns:

(1) Figure 1a uses Ct value to measure Syt1 and Syt7 expression levels, but a comparison with GAPDH Ct cycle numbers in different cell types will be helpful for understanding.

We appreciate the reviewer’s suggestion of including GAPDH Ct cycle numbers. We have revised Figure 1a to show Ct values for Syt1, Syt7, and GAPDH in both BEAS-2B and THP-1 cells.

(2) Figure 1b indel, shown with an ICE method, should be confirmed with protein expression levels to interpret functional results.

We thank the reviewer for raising this concern. We attempted to assess protein levels by western blot using multiple antibodies from both Abcam and Synaptic Systems. However, we were unable to identify a suitable antibody that reliably detected endogenous Syt1 or Syt7 protein levels.

(3) Figure 1c. HLA-B45-restricted T cell clones also show some marginal reduction of IFN-γ spot responses and are more different in Figure 6b. Please discuss this conflicting data. Also, need a reference to support whether the exogenous CFP peptide antigen is presented via surface or endocytic antigen loading.

We agree with the reviewer that there are some marginal reductions of IFN-γ responses for HLA-B45-restricted T cell clones. Since T cell clones are used from frozen, there can be

differences in maximal responses between T cell clones and expansions of the same T cell clone. However, the comparisons include a control arm and pool data from multiple experiments to reach statistical power and validity. In addition, Figure 6b shows Syt1 and Syt7 KO cells in the background of BEAS-2B MR1KO:tetMR1-GFP clone D4 cells, which overexpresses MR1 that may contribute to variability and potentially account for the observed differences. With respect to exogenous CFP peptide loading, earlier studies on peptides and antigen presenting cells demonstrated that peptides can be loaded onto fixed cells and subsequently presented to T cells (Shimonkevitz et al., 1983; Watts et al., 1985). Based on these findings, it is reasonable to assume that substantial peptide exchange occurs at the cell surface when exogenous peptides are added to antigen presenting cells.

(4) Figure 2e: Delta CT values of Syt1, Syt7 in WT, KO cells can be shown together with Ct values of GAPDH or B2m house-keeping genes to help readers determine the efficiency of Syt1 and 7 mutation at the gene expression level. Also, in Figure 4a, the baseline of Ct values for GAPDH can be plotted together.

As suggested by the reviewer, we have revised Figure 2e and 4a to include CT values for the genes of interest as well as housekeeping gene GAPDH.

(5) Figure 3c and Figure 1d: M.smeg infection can be shown to be more comparable with Mtb infection.

We thank the reviewer for this thoughtful comment. Although *M. smegmatis* infection could serve as a comparable control, *M. smegmatis* secretes large amounts of MR1 ligands derived from riboflavin metabolism. This makes it difficult to distinguish between extracellular and intracellular antigens, and to directly compare with Mtb infection, which is specifically an intracellular infection model.

(6) Figure 4e: It appears Esyt2 Knockdown shows strong inhibition of MAIT activation mediated by BEAS2B cells with Mtb infection and M.smeg supernatant stimulation. Please add other relevant data, such as MR1 cell surface expression and colocalization, and discuss these results with Syt proteins.

We appreciate the reviewer's suggestion to include relevant data for Esyt2 knockdown. We performed flow cytometry analysis of Esyt2 knockdown cells and found surface MR1 expression under basal conditions. Treatment with Ac-6-FP resulted in increased MR1 surface stabilization, but MR1 surface level was significantly lower than those observed in missense control cells. Therefore, Esyt2 is not specific to MR1 presentation of Mtb-derived metabolites and instead may play a broader role in overall MR1 antigen presentation, including intracellular Mtb-derived antigens, exogenous antigens, and ER-loaded Ac-6-FP.

(7) Figure 5 colocalization computational analyses can be more explicitly presented regarding randomization, technical procedures, and statistical analyses, as stated in Concern 2.

As suggested, we have included more details in methods section and added the supplemental data.

(8) Figure 6a: Syt1 and Syt7 protein expressions are also suggested to confirm the mutation, similar to the confirmation for Figures 1 and 3.

We thank the reviewer for raising this concern. As discussed previously, we have not identified a suitable antibody for human Syt1 and Syt7. We have tested multiple antibodies from Abcam and Synaptic Systems.

(9) For statistical analyses, "non-linear regression analysis comparing best-fit values of top and EC50 were used to calculate p-values by extra sum-of-squares F test" (Figure 6b)

and "non-linear regression analysis of pairwise comparison to WT on best-fit values of top and EC₅₀ were used to calculate p-values by extra sum-of-squares F test." (Figure 3bc), readers may need more specific demonstration in supplemental figures on how statistical analyses have been performed.

We appreciate the reviewer's suggestion to include more detailed information regarding the statistical analyses. For clarification, data presented in Figures 6b and 3bc were analyzed using the same statistical analysis in Prism 10. Specifically, nonlinear regression (curve fit) was performed using the [Agonist] vs. response model with three parameters. Best-fit values for the top and EC₅₀ parameters were compared using an extra sum-of-squares F test. No constraints were applied to the bottom and top parameters, and the EC₅₀ parameter was constrained to be greater than 0 for p-value calculation. We have revised the Statistical Analysis section of the Methods to more clearly describe this approach.

(10) In discussion, the background section for Syt1 and Syt7 is more appropriate to be in the introduction. This will allow readers to better understand the association of Syt proteins with MR1 and the necessity to study the impact of Syt on MR1 trafficking.

We thank the reviewer for this suggestion. We believe that the basic background and relevance of Syt1 and Syt7 in MR1 trafficking are covered in the introduction; however, we have added details to help readers understand their impact.

Reviewer #2 (Recommendations for the authors):

This reviewer has no requests for implementation and congratulates the authors on this nice piece of work.

We thank the reviewer for the positive comments.

Reviewer #3 (Recommendations for the authors):

Complete trafficking experiments to pinpoint the trafficking relationship between Syt 1 and 7 and MR1 in Mtb infection.

We appreciate the reviewer's insightful comment. As this study represents the first detailed investigation into the roles of Syt1 and Syt7 in MR1-mediated presentation of Mtb-derived metabolites, we agree that a fully resolved trafficking mechanism has not yet been established. A major limitation is the inability to directly track Mtb-derived antigens as they are loaded onto MR1 and trafficked to the cell surface. Therefore, we relied on complementary functional and microscopy-based approaches, including IFN- γ ELISpot assays, flow cytometry, fluorescence microscopy, and flow organellometry, to infer the trafficking relationships between Syt1, Syt7, and MR1 during intracellular Mtb infection. Our data support a model that Syt1 and Syt7 facilitates the trafficking of MR1 from Mtb-containing vacuoles to the plasma membrane. This interpretation is supported with the increased accumulation of MR1 in Mtb-containing vacuoles and reduction in MAIT cell IFN- γ release observed in Syt1 and Syt7 knockout cells.

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